

FINITE FAILURE SCHEMES AND THE RECONSTRUCTION OF QUADRATIC PENCILS

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ABSTRACT. We reconstruct a quadratic pencil varying over a smooth connected projective complex base from an abstract unmarked finite thickening. The scheme recovers the reduced base, the actual source bundle, and the whole pencil subbundle up to one constant left transformation. The theorem holds for every pencil, including singular pencils, and for arbitrary source bundles. It follows from a coefficient-descent principle for moving subspaces in multiplicity-one Schur modules: the first relation kernel recovers a determinant cone, asymmetric coefficient supports orient its tensor factors, and the actual normal map removes the line-twist ambiguity. We distinguish this orbit rigidity from the general first-relation truncation lemma, whose entire nonlinear freedom is a unipotent automorphism kernel. For embedding dimension $n \geq 3$, a single unmarked Artin local algebra already determines a constant pencil at the smallest uniform order $d = n^2 + 2n - 4$; all lower orders forget it. The same order works on the socle Grassmannian or on a Schubert line. We give the local automorphism group as an affine group scheme and retain the filtered geometric automorphism sequence for projective reductions. Two isotrivial families with identical graded vector bundles are separated by their unmarked algebra structures. A nonisotrivial pencil family passes through spectral collisions without loss of reconstruction. The marked universal construction commutes with arbitrary complex base change, and regular pencils retain their full spectral and higher Fitting data.

1. INTRODUCTION

A degeneracy locus usually forgets the map that defines it. We study an inverse problem in which neither the ambient embedding nor the normal directions are retained: can the abstract nonreduced failure scheme recover a quadratic pencil? The answer is affirmative for every complex pencil, including singular pencils. More precisely, there is a smallest uniform infinitesimal order at which the pencil appears. The inverse is also global: a thickening over a projective reduction recovers an entire varying pencil subbundle and its actual source bundle, up to one constant left transformation. No fibrewise choice of coordinates or projective line-bundle ambiguity is left. A single unmarked local algebra suffices for the constant-pencil problem; a Schubert line gives another geometric realization.

Let V be a complex vector space of dimension $n \geq 3$, and put

$$W = \mathrm{Sym}^2 V, \quad N = \binom{n+1}{2}, \quad R \in \mathrm{Gr}(2, W), \quad S_R = W/R, \quad p = N - 2.$$

The cube-zero algebra $A_R = \mathbb{C} \oplus V \oplus S_R$ has multiplication $\mathrm{Sym}^2 V \rightarrow S_R$, with $(V \oplus S_R)S_R = 0$. On $X_R = \mathrm{Gr}(n, V \oplus S_R)$, let $\mathcal{K}$ denote the tautological bundle. For any $m \geq 2$, set

$$\widehat{D}_R = V(\mathrm{Fitt}_0 \mathrm{coker}[\mathrm{Sym}^m(\mathcal{O} \oplus \mathcal{K}) \rightarrow A_R \otimes \mathcal{O}]). \quad (1.1)$$

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This is the maximal-minor scheme of multiplication, not merely its support. It is independent of $m \geq 2$. Its reduction is the Schubert divisor where $\mathcal{K} \rightarrow V$ drops rank. Write $B_R = \text{Gr}(n, S_R)$, let $\mathfrak{b}_R$ be its ideal in $\widehat{D}_R$, and define

$$Z_R^{[k]} = V(\mathfrak{b}_R^{k+1}) \subset \widehat{D}_R, \quad d = n + 2p = n^2 + 2n - 4.$$

An isomorphism of these neighbourhoods means an abstract complex scheme isomorphism; it is supplied with no ambient or tensor marking.

Theorem 1.1 (Sharp intrinsic reconstruction). *For all quadratic pencils $R, R' \subset \text{Sym}^2 V$,*

$$Z_R^{[d]} \simeq Z_{R'}^{[d]} \iff R' = gR \text{ for some } g \in \text{PGL}(V).$$

For every $0 \leq k < d$, the isomorphism class of $Z_R^{[k]}$ is independent of R . Thus d is the smallest uniform reconstruction order. The same statements hold after restriction, using the graph projection, to any Schubert line in B_R . The resulting abstract curve-supported neighbourhood is independent of the choice of that line.

The Grassmannian statement is Theorem 7.1. The curve-supported statement is Corollary 8.2. Its more general form, Theorem 8.1, applies over any smooth connected projective curve with a rank- n bundle having nontrivial projectivization. For example, $\mathcal{O}_C(-P) \oplus \mathcal{O}_C^{n-1}$ works on every such pointed curve. The curve is intrinsic as the reduction; a chosen embedding of the curve is not part of the isomorphism data.

1.1. A moving family, not only its individual fibres. The same finite object also reconstructs a pencil that varies over its projective reduction. Let B be a smooth connected projective complex variety, let $\mathcal{U}$ have rank n , and let $\mathcal{R}$ be a rank-two subbundle of $\text{Sym}^2 V \otimes \mathcal{O}_B$ with locally free quotient. Use the local ideal $(\det T)I_p(\gamma \text{Sym}^2 T)$ in $\text{Hom}(\mathcal{U}, V)$ and truncate at order d . Theorem 10.2 states that two resulting abstract finite schemes are isomorphic exactly when their data are related by an isomorphism of reduced bases, one constant linear left map, and an actual isomorphism of the source bundles, carrying one pencil subbundle to the other. The projection and both tensor factors are forgotten in the input. The varying coefficient line is recovered as a subbundle, not by pointwise choices of projective coordinates.

This strengthens the constant-pencil curve theorem in two directions: the reduced base and source bundle may vary in the comparison, and the pencil need not be constant on that base. Example 9.6 has a nonconstant unordered cross-ratio and six spectral collision points, all retained by the same finite scheme. Theorem 9.4 then identifies the remaining freedom in an isomorphism. Its quotient is the group of the recovered linear data; its kernel acts trivially on the associated graded scheme and has a terminating commutator filtration. Explicit nonlinear shears show why that kernel cannot be omitted. These statements are consequences of the unmarked inverse, not additional claims of novelty for the classical Pluecker map or for spectral cross-ratios.

1.2. A single unmarked local algebra suffices. The strongest form of the inverse needs no positive-dimensional support. Choose an n -dimensional source space U , write $T = (t_{ij})$ for its universal map to V , and let $\mathfrak{m} = (t_{ij})$. Define

$$\mathfrak{A}_R^{[d]} = \mathbb{C}[t_{ij}] / ((\det T)I_p(\gamma_R \text{Sym}^2 T) + \mathfrak{m}^{d+1}).$$

Theorem 1.2 (Local algebraic inverse). *For every pair of complex quadratic pencils, an ungraded algebra isomorphism $\mathfrak{A}_R^{[d]} \simeq \mathfrak{A}_{R'}^{[d]}$ exists if and only if $R' = gR$ for some $g \in \text{PGL}(V)$. The order $d = n^2 + 2n - 4$ is the smallest uniform order already for these single-point objects.*

The full statement and proof are Theorem 10.3. Proposition 2.1 separates the general truncation argument from the pencil-specific orbit rigidity: nonlinear coordinate changes form an explicitly described unipotent kernel, whereas the first relation space carries the reconstruction data. The extra local ingredient is that the residual coefficient module has factor-support ranks $(1, \binom{N}{2})$. Transposition reverses these unequal ranks and therefore cannot carry one pencil ideal to another. This orients the unordered matrix factors without a nontrivial projective bundle. Theorem 10.2 consequently removes that restriction from the moving-family theorem as well. This is not a claim that the reduced determinant cone alone remembers the pencil: the nonreduced first relation kernel and its residual coefficients are indispensable. Proposition 10.4 also identifies the entire tangent-identity kernel, including its affine-space parametrization and nonadditive group law.

1.3. The intrinsic mechanism. In the graph neighbourhood of B_R , with tautological bundle $\mathcal{U}$ and graph map $T : \mathcal{U} \rightarrow V$, the ideal is

$$(\det T)I_p(\gamma_R \operatorname{Sym}^2 T).$$

Its generators have degree d . This degree calculation alone only explains why lower orders are constant. The inverse requires three further intrinsic operations. Multiplication in the finite scheme recovers the first relation kernel and hence the whole homogeneous cone. The rank-one Fano scheme of its reduced determinant cone then recovers the two tensor rulings. On the socle Grassmannian one may distinguish them by projective triviality, already on a Schubert line. In general the unequal coefficient-support ranks orient them without that restriction. Properness forces the induced map to $\operatorname{PGL}(V)$ to be constant. After choosing its linear lift, the actual normal-bundle map lies in the commutant of $\operatorname{End}(V)$, which recovers the right bundle without a line twist (Lemma 4.2). Finally, cancellation of the intrinsic determinant ideal recovers the coefficient module. For pencils its exterior representation is irreducible, and exterior duality identifies its single coefficient line with the Pluecker line of R .

Theorem 5.1 separates this geometric mechanism from the particular multiplication algebra. The curve theorem shows why the mechanism is not tied to the dimension of the original parameter Grassmannian. The geometric orientation used here has a local replacement: coefficient-support asymmetry. Theorem 10.3 uses it to reconstruct the pencil from one unmarked Artin local algebra.

Theorem 4.3 gives the broader principle. It reconstructs simultaneously varying subspaces in several Schur modules from the unmarked first-relation thickening, provided one coefficient support is nonzero and proper. The coefficient spaces may have arbitrary prescribed ranks; they need not be lines coming from quadratic pencils. Its proof isolates the actual factor-descent step from classical linear-preserver theory. Example 4.4 shows why the global descent is not equivalent to fibrewise reconstruction: two Zariski locally trivial families have the same Artin fibre and isomorphic nilradical graded vector bundles in every degree, but their abstract total schemes differ. The inverse detects the inequivalent degree-three coefficient maps. This is the mechanism behind the moving-pencil theorem, rather than an assertion of novelty for canonical grading or for the numerical value of the first relation degree.

1.4. Framed families and spectral consequences. Once tensor coordinates are retained, the parameter map is simpler. The first relation spaces give a closed immersion of $\operatorname{Gr}(2, W)$ into a Grassmannian, by Theorem 7.3. Its proof factors through the classical Pluecker embedding, tensoring a line with a fixed space, and a fixed linear inclusion. This is the framed encoding consequence of the coefficient calculation, not a substitute for the unmarked inverse. Theorem 12.1 glues the actual universal order- d neighbourhood. It is finite locally free over the relative socle of rank $\binom{n^2+d}{d} - \binom{N}{2}$, and projective and flat over the pencil parameter scheme. It commutes with arbitrary complex base change with that socle retained. Over a nonreduced base, the specified

relative socle ideal need not be the absolute nilradical. These statements do not identify unmarked moduli stacks or deformation groupoids.

For regular pencils, the spectral sheaf and its Fitting incidence schemes are read from the reconstructed pencil. We retain these consequences because they exhibit information genuinely lost by coarser invariants: fixed discriminants and fixed reduced rank loci can conceal different elementary divisors. In Theorem 13.2, every classical dominance specialization is transported to a flat family of the finite neighbourhoods, with every strict transition detected at order d . The spectral orbit classification itself is classical, not a second independent Torelli theorem.

1.5. Prior work and the publication object. The Weierstrass–Segre classification is used in its root-labelled form; compare [4, Theorem 1.1 and Corollary 2.1]. The orthogonal nilpotent input is isolated in Lemma 13.1, with precise references to [5, Proposition 1 and Theorem 1], an independent dimension calculation, and an explicit distinction between full orthogonal orbits and their special-orthogonal components.

Ballico’s relative multiplication construction [2, Theorem 0.2; §1, Remark 1.2 and equation (6)] is a substantive antecedent for studying quadratic-normality failure on finite linear sections of an embedded surface. Here the defining algebra varies, and the inverse starts from an unmarked infinitesimal failure scheme. The opening page of the distinct earlier paper [1, p. 5] concerns higher-order embedding properties and restriction maps to finite subschemes. Its remaining theorem and proof text has not been inspected; that first page does not establish the absence of an inverse statement. No anticipation or nonanticipation conclusion relative to that article is asserted.

The Artin-local aspect is compared at theorem level with [7, 8] in Section 2, including the distinction between canonical grading of a nonhomogeneous local algebra and orbit rigidity of our already homogeneous first relation. The linear-preserver input is classical [9]; its precise hypotheses and a proof in the required case are included in Lemma 4.1.

The present article is the reconstruction argument and its spectral readout. The reciprocal-likelihood theorem and projective critical schemes are preserved in a separate applications manuscript. Its finite-flat critical divisor requires supplied split semisimple data and simple disjoint projective poles; its totally real cover requires a supplied real definite realization and separated data. None of those hypotheses enters the all-pencil theorem here. Independent web, exterior-operator, K3, polarized, and boundary results are preserved in an explicitly non-submitted research archive. They are not inputs to any proof in this article and are not part of its significance claim. Every proof required here is included here.

2. FIRST RELATIONS, UNGRADED ISOMORPHISMS, AND INVERSE SYSTEMS

The passage from a finite local algebra to its first relation space is independent of pencils. We isolate it before undertaking the geometric inverse. In this section E denotes a space of degree-one *generators*; thus it is a cotangent space, rather than the dual normal space used later. Let $e = \dim E$, $d \geq 2$, and $K \subset \mathrm{Sym}^d E$. Set

$$A(K) = \mathrm{Sym} E / ((K) + \mathfrak{m}^{d+1}), \quad \mathfrak{m} = (E), \quad \mathfrak{n} = \mathfrak{m}A(K). \quad (2.1)$$

The grading in this presentation is not supplied to an isomorphism.

Proposition 2.1 (The first-relation orbit principle). *The abstract algebra $A(K)$ recovers E and K , up to linear isomorphism, by*

$$E = \mathfrak{n}/\mathfrak{n}^2, \quad K = \ker[\mathrm{Sym}^d(\mathfrak{n}/\mathfrak{n}^2) \longrightarrow \mathfrak{n}^d]. \quad (2.2)$$

Consequently $A(K) \simeq A(K')$ as ungraded complex algebras if and only if some $a : E \xrightarrow{\sim} E'$ satisfies $(\mathrm{Sym}^d a)K = K'$.

More precisely, after fixing the homogeneous inclusions of generators, all isomorphisms with tangent map a are exactly the substitutions

$$x \mapsto a(x) + u(x), \quad u \in \operatorname{Hom}_{\mathbb{C}}(E, \mathfrak{n}^2).$$

These assertions about substitutions remain valid after scalar extension to any commutative complex algebra.

Proof. The positive ideal is the unique maximal ideal, and $\mathfrak{n}^{d+1} = 0$. Multiplication of d lifts of cotangent vectors is independent of their lifts: replacing any one by an element of $\mathfrak{n}^2$ changes the product in $\mathfrak{n}^{d+1}$. There are no relations of smaller degree, and the degree- d quotient is $\operatorname{Sym}^d E/K$. This proves (2.2). In particular every isomorphism induces the stated linear equivalence.

Conversely substitute $a(x) + u(x)$ for each generator. A relation of degree d changes, beyond its linear substitution, only in degrees at least $d + 1$. Those terms vanish. The relations of degree $d + 1$ also vanish. The induced map on associated graded algebras is the isomorphism induced by a ; a finite filtration then makes the algebra map invertible. Every map with this tangent map has precisely such generator images. These polynomial identities hold over any commutative complex coefficient algebra. The intrinsic preservation of the augmentation under that scalar extension is verified in the next proof, so no preservation of a chosen origin has tacitly been added to the isomorphism problem. $\square$

Proposition 2.2 (The full automorphism group scheme). *Write $\ell = \dim_{\mathbb{C}} A(K) = \binom{e+d}{d} - \dim K$ and $H_K = \operatorname{Stab}_{\operatorname{GL}(E)}(K)$, with its closed subgroup scheme structure. The functor $D \mapsto \operatorname{Aut}_D(A(K) \otimes_{\mathbb{C}} D)$ is an affine group scheme of finite type and fits into a split exact sequence of affine group schemes over $\mathbb{C}$:*

$$1 \longrightarrow U_K \longrightarrow \operatorname{Aut}(A(K)) \longrightarrow H_K \longrightarrow 1. \quad (2.3)$$

The kernel is unipotent. Choosing the homogeneous lifting of E identifies its underlying scheme, but not generally its group law, with the affine space associated to $\operatorname{Hom}_{\mathbb{C}}(E, \mathfrak{n}^2)$. Its dimension is $e(\ell - 1 - e)$. Intrinsically, U_K acts simply transitively on the affine space of linear splittings of $\mathfrak{n} \rightarrow \mathfrak{n}/\mathfrak{n}^2$. The splitting of (2.3) uses a choice of such a lifting; it is not an additional intrinsic grading.

Proof. Choose a basis of the finite-dimensional vector space $A = A(K)$. Inside $\operatorname{GL}(A)$, the equations $f(1) = 1$ and $f(xy) = f(x)f(y)$, imposed on basis elements, are polynomial equations. They represent the automorphism functor, including its values on nonreduced complex algebras.

The augmentation can be recovered without mistaking the absolute nilradical after scalar extension for the augmentation ideal. If M_z denotes multiplication by z , the homogeneous filtration gives

$$\operatorname{tr}(M_z) = \ell \epsilon(z). \quad (2.4)$$

Indeed multiplication by a positive-degree element is strictly triangular, whereas multiplication by a scalar has that scalar on every diagonal entry. Identity (2.4) holds over every commutative complex algebra D . Algebra automorphisms conjugate multiplication matrices, so they preserve $\epsilon = \ell^{-1} \operatorname{tr}(M_{(-)})$, its kernel $\mathfrak{n}_D$, and all its powers. Thus the tangent map is a morphism of group schemes into H_K , not merely a map on complex points.

Linear substitutions give a morphism $H_K \rightarrow \operatorname{Aut}(A)$ sectioning this tangent map. By Proposition 2.1, its kernel has, functorially in D , the unique parametrization $x \mapsto x + u(x)$ with $u : E_D \rightarrow \mathfrak{n}_D^2$. Reading generator images and making these substitutions are inverse polynomial maps. On a basis ordered by the positive filtration all kernel matrices are unitriangular, proving unipotence. Multiplication is substitution, which need not be addition of corrections. The dimension follows from $\dim \mathfrak{n}^2 = \ell - 1 - e$.

Finally two splittings of $\mathfrak{n} \twoheadrightarrow E$ differ by exactly a map $E \rightarrow \mathfrak{n}^2$. Any splitting generates A , and its degree- d products impose exactly K ; the preceding substitution argument therefore gives the unique element of U_K relating any two splittings. This is the asserted coordinate-free torsor description. $\square$

2.1. The inverse-system description and its precise role. Put $E^\vee = \text{Hom}_{\mathbb{C}}(E, \mathbb{C})$. Use the apolar pairing in which dual divided monomials satisfy $\langle x^\alpha, y^\beta / \beta! \rangle = \delta_{\alpha\beta}$. Here the pairing means evaluation at the origin after applying the constant-coefficient differential operator. For $I = (K) + \mathfrak{m}^{d+1}$, its Macaulay inverse system is

$$I^\perp = \bigoplus_{j < d} \text{Sym}^j E^\vee \oplus K^\perp, \quad K^\perp \subset \text{Sym}^d E^\vee. \quad (2.5)$$

To check this equality, the relations of degree $d+1$ bound the degree of every inverse-system element by d . The degree- d relations pair only with its top component. All lower components are unrestricted. Conversely that direct sum is stable under every partial derivative: differentiating its top part lands in the full lower-degree space. It is therefore annihilated by I , proving the displayed equality. Thus the orbit problem for K is equivalently the contragredient orbit problem for $K^\perp$. This uses the classical inverse-system correspondence; it does not assert a new canonical-grading theorem.

Elias and Rossi prove that Gorenstein local algebras with Hilbert function $(1, e, e, 1)$ are canonically graded [7, Theorem 3.3 and Corollaries 3.4–3.5]. Their proof removes a lower-degree term of the inverse-system generator by a nonlinear coordinate change. In their later work, compressed Gorenstein algebras of socle degree at most four are canonically graded, with counterexamples beyond the stated range [8, Theorem 3.1 and Examples 3.3–3.5]. In contrast, (2.1) is homogeneous by construction, and its truncation makes all higher substitutions invisible to its first relation. We do not deduce that an arbitrary filtered deformation with the same Hilbert function is graded.

For the pencil algebras, $e = n^2$, $d = n^2 + 2n - 4 \geq 11$, and $\dim K = \binom{N}{2}$. They are not Gorenstein: their socles contain $\mathfrak{n}^d$, whose dimension is $\binom{e+d-1}{d} - \binom{N}{2} > 1$. For example, the first binomial is at least $\binom{e+1}{2}$, while $e = n^2 > N$. The issue left by Proposition 2.1 is accordingly not the removal of a nonhomogeneous inverse-system term. It is the recovery of the matrix factors and defining coefficient data from the *unrestricted* $\text{GL}(E)$ -orbit of K . The next sections establish that recovery and its global descent.

3. INTRINSIC TENSOR RULINGS

The first step of the inverse problem is to recognize the two tensor factors of a determinant cone without an ambient marking. The following scheme-theoretic ruling calculation and descent lemma supply precisely this step. They apply to the pencil argument independently of the web and exterior-contraction constructions in the separate research archive.

Lemma 3.1 (The relative scheme of maximal rank-one spaces). *Let B be any complex scheme and E, F vector bundles of rank $n \geq 2$. In $\text{Gr}_B(n, E \otimes F)$, let $Z(E, F)$ be the zero scheme of the restriction of the quadratic rank-one equations to the universal subbundle $\mathcal{T}$. Equivalently it is the zero scheme of*

$$\text{Sym}^2 \mathcal{T} \longrightarrow \bigwedge^2 E \otimes \bigwedge^2 F.$$

The natural morphism

$$\mathbb{P}_B(E) \sqcup \mathbb{P}_B(F) \longrightarrow Z(E, F), \quad \ell \mapsto \ell \otimes F, \quad m \mapsto E \otimes m, \quad (3.1)$$

is an isomorphism. It commutes with arbitrary change of complex base, including change to a nonreduced base.

Proof. We first work over $\mathbb{C}$. Each displayed map to the Grassmannian is a closed immersion. Its composition with the Pluecker embedding is the degree- n Veronese embedding on $\ell^{\otimes n} \otimes \det F$, respectively $\det E \otimes m^{\otimes n}$, followed by its natural linear inclusion into $\Lambda^n(E \otimes F)$. This linear inclusion is injective: polarization of the nonzero pure-power map gives the corresponding irreducible symmetric-power summand. The rank-one equations vanish identically on both universal families, so the closed immersions factor through $Z(E, F)$.

Every geometric point is in one of the two images. Indeed the sum of two rank-one tensors is rank at most one only if their first factors or their second factors are proportional. Starting with two independent tensors, this observation forces a linear space consisting of rank-one tensors to fix the same factor throughout. A space of dimension n is then exactly one of the displayed spaces. The two images are disjoint: a space fixing both factors has dimension one, not n .

We verify that these pointwise statements omit no infinitesimal structure. At $\ell \otimes F$, choose a generator of ℓ and a complement in E . A Grassmannian tangent vector is a map $\phi : F \rightarrow (E/\ell) \otimes F$. For each coordinate of E/ℓ , write its component as $A : F \rightarrow F$. Linearizing the restricted minors gives

$$u \wedge A(u) = 0 \quad \text{for every } u \in F.$$

Polarizing this diagonal identity gives

$$u \wedge A(v) + v \wedge A(u) = 0 \quad (u, v \in F).$$

In characteristic zero the diagonal and polarized identities are equivalent. These are precisely all coefficients of the bilinear linearization of the restricted quadratic map on $\text{Sym}^2 F$, so no infinitesimal equation is omitted by testing the diagonal. Taking the basis vectors shows that A is diagonal; taking their pairwise sums makes all its diagonal entries equal. Conversely a scalar A satisfies all the linearized equations. Thus the Zariski tangent space to $Z(E, F)$ is precisely $\text{Hom}(\ell, E/\ell)$, of dimension $n - 1$. The other image has the symmetric calculation.

At every closed point the closed immersed projective space gives local dimension at least $n - 1$, whereas the tangent calculation gives embedding dimension exactly $n - 1$. The local ring is therefore regular. A finite-type complex scheme whose closed-point local rings are regular is reduced: a nonempty support of its nilradical would contain a closed point. In particular there is no nilpotent thickening or embedded associated structure at either image. The disjoint union of the two closed immersions is itself a closed immersion, surjective on geometric points. Its ideal in the now reduced target vanishes at every closed point, so it is zero. This proves the scheme isomorphism over $\mathbb{C}$.

For general B , trivialize E, F on a common open cover. The Grassmannian and the zero scheme of the displayed universal quadratic map are, on such an open, the base change of the constant construction just proved. Thus (3.1) is an isomorphism there even if the base has nilpotents. The maps are natural under changes of frame and hence glue. This argument is an identification by equations and base change, not an inference of relative flatness from a fibrewise tangent calculation. $\square$

Lemma 3.2 (Oriented relative Segre descent). *Let B be a connected projective complex variety with $H^0(B, \mathcal{O}_B) = \mathbb{C}$, and let V, V' have the same dimension $n \geq 2$. Let $\mathcal{U}, \mathcal{U}'$ be vector bundles of rank n on B . Suppose a linear bundle isomorphism*

$$\alpha : V \otimes \mathcal{U}^* \xrightarrow{\sim} V' \otimes \mathcal{U}'^*$$

preserves the rank-one cones and the ruling whose parameter bundle is $\mathbb{P}(V) \times B$. The induced isomorphism $\mathbb{P}(V) \times B \simeq \mathbb{P}(V') \times B$ is regular and is induced by one constant projective isomorphism

$[g] : \mathbb{P}(V) \rightarrow \mathbb{P}(V')$. Locally $\alpha(T) = gTh^{-1}$, with the same $[g]$ on every chart. The possible line-bundle twists of vector-bundle lifts do not change the projective left coefficient lines of any homogeneous functorial coefficient construction.

Proof. Lemma 3.1, applied to $E = V \otimes \mathcal{O}_B$ and $F = \mathcal{U}^*$, identifies the closed relative Fano scheme with $(\mathbb{P}(V) \times B) \sqcup \mathbb{P}_B(\mathcal{U}^*)$. In particular its two components carry their displayed scheme structures, with no additional nilpotent or embedded components. The isomorphism α induces a regular map of these relative parameter schemes, and the orientation hypothesis preserves the first component.

For completeness, an isomorphism of two trivial $\mathbb{P}^{n-1}$ -bundles over any base gives a regular projective-matrix-valued map, not merely a fibrewise collection. Its pullback of the relative $\mathcal{O}(1)$ has the form $\mathcal{O}(1) \otimes p^*L$ for a line bundle L on the base. To see this, tensor by $\mathcal{O}(-1)$; the resulting line bundle is trivial on every projective-space fibre, and its pushforward and evaluation identify it with a pullback from the base. Pushing forward $\mathcal{O}(1)$ now gives an isomorphism of the corresponding trivial rank- n bundles, with that line-bundle twist. On charts trivializing L , it is an invertible matrix g_i of regular functions. On overlaps, $g_j = u_{ij}g_i$ for an invertible regular function u_{ij} . The classes $[g_i]$ thus glue to a regular morphism to the projective-isomorphism scheme, identified after choosing fixed bases with PGL_n . This accounts explicitly for the possible absence of a global linear lift.

The group PGL_n is affine: it is the determinant-nonzero principal open in the projective space of matrices. A morphism from B to an affine scheme factors through $\mathrm{Spec} H^0(B, \mathcal{O}_B) = \mathrm{Spec} \mathbb{C}$, so the projective map is constant. Choose a constant linear lift g . Each local g_i then differs from g by a scalar unit. After removing its two projective-factor actions, a linear map preserving every decomposable line is scalar: apply it to sums of two tensors with one common factor. Consequently the remaining action is a right bundle map, and the scalar units may be absorbed in its local lifts h . This proves the local tensor formula.

Finally a scalar multiplying a lift acts on a homogeneous functor of degree a by its scalar to the power a (or $-a$ on its dual). It cannot alter a projective coefficient line. The conclusion applies to every homogeneous degree. In particular, no right change of frame can replace the single projective left map by separate transformations on different coefficient spaces. $\square$

4. ACTUAL FACTOR DESCENT AND MOVING COEFFICIENT SPACES

We now separate the linear-preserver input, the descent of vector bundles, and the recovery of coefficients. The resulting principle allows several varying coefficient spaces, of arbitrary prescribed ranks, rather than only the line associated with a pencil.

Lemma 4.1 (Linear preservers of the determinant cone). *Let U, V, U', V' be complex vector spaces of dimension $n \geq 2$. An invertible linear map $\Phi : \mathrm{Hom}(U, V) \rightarrow \mathrm{Hom}(U', V')$ carrying the determinant hypersurface onto the determinant hypersurface has one of the forms*

$$T \mapsto gTh^{-1}, \quad T \mapsto gT^{\mathfrak{t}}h^{-1},$$

where the second formula is written after choosing identifications for transposition. The two cases respectively preserve and exchange the two rank-one rulings. In the first case the pair (g, h) is unique up to a common nonzero scalar.

Proof. At a matrix of rank r , an invertible r -minor reduces the local rank condition to the rank condition on a generic $(n - r)$ -square matrix. The singular locus of the rank-at-most s variety is therefore the rank-at-most $s - 1$ variety for $1 \leq s < n$: on the rank- s stratum the Schur-complement entries give independent equations, whereas on a lower stratum the linear terms of its defining $(s + 1)$ -minors vanish. Starting with the determinant cone and iterating

reduced singular loci hence recovers the rank-one cone. For $n = 2$ that cone is already the determinant cone.

If $v \otimes u$ and $w \otimes z$ have rank one, their sum has rank at most one only if v, w or u, z are dependent. It follows that each maximal linear space of rank-one tensors has one factor fixed. The two families are $\mathbb{P}(V)$ and $\mathbb{P}(U^*)$. The induced Segre automorphism must preserve these families or exchange them. In the preserving case its restriction to a ruling gives projective linear maps on each factor. Remove linear lifts of both maps. The remaining linear transformation fixes each decomposable line. In tensor bases its eigenvalues at $v_i \otimes u_j$ agree along each row and each column by applying it to $(v_i + v_k) \otimes u_j$ and $v_i \otimes (u_j + u_l)$. They are all equal, so the remaining map is scalar. This proves the first form and its uniqueness. Composing with transposition reduces the exchanging case to the first.

This is the square invertible case of the classical rank-one preserver theorem of Marcus and Moyls [9, Theorem 1, pp. 1218–1219]; the argument above specifies the reduction from the determinant hypersurface that is needed here. $\square$

Lemma 4.2 (An actual tensor map removes the line twist). *Let B be a reduced complex variety, V a complex vector space with $\dim V \geq 2$, and $\mathcal{A}_1, \mathcal{A}_2$ vector bundles. Suppose an actual bundle isomorphism*

$$\Psi : V \otimes \mathcal{A}_1 \xrightarrow{\sim} V \otimes \mathcal{A}_2$$

preserves the rank-one cones and induces the identity on the $\mathbb{P}(V)$ -parameter space of rulings. There is a unique bundle isomorphism $a : \mathcal{A}_1 \rightarrow \mathcal{A}_2$ with $\Psi = 1_V \otimes a$.

Equivalently, the intrinsic operation of taking the commutant of the constant matrix algebra gives

$$\mathcal{H}om_{\text{End}(V) \otimes \mathcal{O}_B}(V \otimes \mathcal{A}_1, V \otimes \mathcal{A}_2) \simeq \mathcal{H}om_{\mathcal{O}_B}(\mathcal{A}_1, \mathcal{A}_2). \quad (4.1)$$

Thus after a constant left projective map has been lifted linearly, no residual line-bundle twist is left in the recovery of the right factor from Ψ .

Proof. For every constant line $\mathbb{C}v \subset V$, the corresponding ruling is fixed, so on every geometric fibre Ψ maps $v \otimes \mathcal{A}_1$ onto $v \otimes \mathcal{A}_2$. Choose a basis $v_1, \dots, v_n$ for the purpose of checking this assertion as an equality of bundle maps. The rulings for v_i force all off-diagonal blocks of Ψ to vanish. The rulings for $v_i + v_j$ force its diagonal blocks to agree. Since B is reduced, fibrewise zero morphisms of vector bundles are zero. Therefore Ψ commutes with every constant matrix unit, hence with $\text{End}(V) \otimes \mathcal{O}_B$.

For clarity, commutant recovery itself is canonical and does not use these chosen matrix units as extra data. Tensoring a bundle map a with 1_V gives the forward map in (4.1). Commutation with matrix units shows locally that it is bijective, with inverse reading any diagonal block. This proves that it is an isomorphism of sheaves, independent of the checking basis. Its inverse applied to the global section Ψ produces a global section a , not projective local sections requiring arbitrary scalar lifts. Applying the same operation to Ψ^{-1} proves that a is invertible. Uniqueness follows from injectivity of $a \mapsto 1_V \otimes a$.

Projective-factor data alone would permit replacing a bundle by a line twist. The additional datum used here is the actual linear map of the tensor bundles. It is this datum, recovered from the dual of the first nilradical quotient, that makes the commutant argument applicable. $\square$

4.1. A general finite-order coefficient inverse. Fix $n \geq 2$, $q \geq 1$, and integers $0 \leq r_\lambda \leq m_\lambda = \dim \mathbb{S}_\lambda \mathbb{C}^n$ for partitions λ of q of length at most n . Assume that for at least one partition

$$0 < r_\lambda < m_\lambda. \quad (4.2)$$

Let B be a smooth connected projective complex variety, V a constant n -dimensional space, and $\mathcal{U}$ a rank- n bundle. For each λ , choose a subbundle with locally free quotient

$$\mathcal{L}_\lambda \subset \mathbb{S}_\lambda V^* \otimes \mathcal{O}_B, \quad \text{rank } \mathcal{L}_\lambda = r_\lambda.$$

Zero and full subbundles are permitted. In the matrix bundle $\mathcal{E} = \text{Hom}(\mathcal{U}, V \otimes \mathcal{O}_B)$, the multiplicity-one Cauchy decomposition supplies the subbundle

$$\mathcal{J}_q = \bigoplus_{\lambda} \mathcal{L}_\lambda \otimes \mathbb{S}_\lambda \mathcal{U} \subset \text{Sym}^q \mathcal{E}^*. \quad (4.3)$$

Let $\mathcal{D}_n \subset \text{Sym}^n \mathcal{E}^*$ be the determinant line, put $D = n + q$, and define

$$\mathcal{K} = \mathcal{D}_n \mathcal{J}_q, \quad \mathcal{A} = \text{Sym } \mathcal{E}^* / ((\mathcal{K}) + \mathfrak{a}^{D+1}), \quad Y = \text{Spec}_B \mathcal{A}, \quad (4.4)$$

where $\mathfrak{a}$ is the positive homogeneous ideal. The notation D in this subsection is an integer, not a failure locus. The scheme Y is finite over B , and need not be zero-dimensional over $\mathbb{C}$.

Theorem 4.3 (Unmarked descent for moving coefficient spaces). *For fixed $n, q, (r_\lambda)$ satisfying (4.2), the abstract complex scheme Y recovers its coefficient data as follows. Two such schemes Y_1, Y_2 are isomorphic if and only if there are an isomorphism $\sigma : B_1 \rightarrow B_2$, a constant linear isomorphism $g : V_1 \rightarrow V_2$, and an actual bundle isomorphism $h : \mathcal{U}_1 \rightarrow \sigma^* \mathcal{U}_2$ such that, for every λ ,*

$$\mathcal{L}_{1,\lambda} = (\mathbb{S}_\lambda g)^* \sigma^* \mathcal{L}_{2,\lambda}. \quad (4.5)$$

Every unmarked isomorphism induces a normal-bundle map $T \mapsto gTh^{-1}$; for that map the pair (g, h) is unique up to a common nonzero constant scalar. Conversely every such triple lifts homogeneously to an isomorphism. The scheme is finite locally free over its reduced base of rank

$$\binom{n^2 + D}{D} - \sum_{\lambda} r_\lambda m_\lambda.$$

No nontriviality hypothesis on $\mathbb{P}_B(\mathcal{U}^)$ is required.*

Proof. Recovering the intrinsic linear data. Multiplication by a determinant polynomial is an injective linear map of polynomial spaces. In local bundle frames its matrix is constant over $\mathbb{C}$ and its cokernel is free. The Cauchy summands and the $\mathcal{L}_\lambda$ are subbundles, so $\mathcal{K}$ is a subbundle of rank $\sum r_\lambda m_\lambda$. The graded pieces of $\mathcal{A}$ below D are the full symmetric powers; its top piece is $\text{Sym}^D \mathcal{E}^* / \mathcal{K}$. This proves local freeness and the rank formula.

The nilradical $\mathfrak{n}$ of $\mathcal{O}_Y$ is its positive ideal, since B is reduced. Thus $Y_{\text{red}} = B$ and $\mathfrak{n}/\mathfrak{n}^2 = \mathcal{E}^*$ intrinsically. The multiplication kernel

$$\ker[\text{Sym}^D(\mathfrak{n}/\mathfrak{n}^2) \rightarrow \mathfrak{n}^D]$$

is precisely $\mathcal{K}$, by the proof of Proposition 2.1 in local frames. An abstract isomorphism consequently gives the base isomorphism σ and an actual linear normal-bundle isomorphism carrying $\mathcal{K}_1$ to $\mathcal{K}_2$ contravariantly. The normal map is linear over the reduced base even when the original scheme map does not preserve the chosen graph projection. Indeed the action on $\mathfrak{n}/\mathfrak{n}^2$ is linear over $\mathcal{O}_Y/\mathfrak{n}$.

Recovering and orienting the factors. Generate the homogeneous ideal $(\mathcal{K})$ in the recovered symmetric algebra; do not confuse its cone with the finite truncation. Its radical is the determinant ideal. To verify this, at an invertible matrix T , the map $\mathbb{S}_\lambda T$ is invertible. For a partition with $r_\lambda > 0$, a nonzero covector in $\mathcal{L}_\lambda$ has a nonzero matrix coefficient there. Thus the residual coefficients cannot all vanish off the determinant. All generators vanish on the determinant. The relative determinant hypersurface is reduced over the smooth base; the equality of radicals follows on affine charts from this equality of closed points. Successive reduced singular loci and Lemma 3.1 recover its two unordered tensor rulings with their relative scheme structures.

Cancellation of the determinant line is intrinsic and recovers the ideal generated by $\mathcal{J}_q$, and hence its degree- q piece. In a Cauchy summand the two contraction supports of $\mathcal{L}_\lambda \otimes \mathbb{S}_\lambda \mathcal{U}$ have ranks (r_λ, m_λ) . Separate factor changes preserve these ranks. By Lemma 4.1, an exchange of the two matrix factors interchanges the ranks in each summand. At a partition satisfying (4.2), the result (m_λ, r_λ) cannot equal (r_λ, m_λ) for the other datum. Thus no fibre permits an exchange. Twisting local factor lifts by lines changes no support rank. The V -ruling is thereby oriented intrinsically, including when both projective ruling bundles are trivial.

Descending the actual map. By Lemma 3.2, the induced morphism $B_1 \rightarrow \text{Isom}(\mathbb{P}(V_1), \mathbb{P}(V_2))$ is constant: the target is affine and $H^0(B_1, \mathcal{O}_{B_1}) = \mathbb{C}$. Choose a constant linear lift g . Remove it from the actual normal map. Lemma 4.2, applied to $\mathcal{A}_1 = \mathcal{U}_1^*$ and $\mathcal{A}_2 = (\sigma^* \mathcal{U}_2)^*$, supplies the unique global right factor. Dualizing its inverse gives the actual isomorphism h . This step uses more than an isomorphism of projective bundles; it is exactly what excludes an undetected line twist. Replacing g by cg replaces h by ch . These are all ambiguities because the left lift is constant.

Recovering every coefficient subbundle. The oriented Cauchy decomposition is now functorial for the recovered left-right map. Project the residual degree- q module onto each summand and contract against the full right dual factor. The image is precisely $\mathcal{L}_\lambda$, as a subbundle of the constant left module, not just a collection of fibrewise projective subspaces. The right map acts invertibly on the full Schur factor and hence does not change this image. Pullback by the normal map therefore gives (4.5) for the same constant g in all summands. Scalar choices in the determinant line do not change these subbundles. This proves necessity.

Conversely $T \mapsto gTh^{-1}$ carries the determinant line, all coefficient subbundles, and every power of the positive ideal to the corresponding objects. It induces the required homogeneous scheme isomorphism. The stated uniqueness refers to the induced normal map; it does not eliminate the higher-order automorphisms of that scheme. $\square$

4.2. Identical fibres and graded bundles do not determine a family. The global conclusion has content even when neither fibre isomorphism classes nor the underlying graded vector bundles vary. The following example uses the general coefficient theorem, not a further spectral readout of a reconstructed pencil.

Example 4.4 (Two isotrivial thickenings separated by coefficient descent). Take $B = \mathbb{P}^1$, $V = \mathbb{C}^3$, $\mathcal{U} = \mathcal{O}_B^3$, $q = 1$, and $r_{(1)} = 1$. The two everywhere nonzero columns

$$(s^3, t^3, 0)^t, \quad (s^3 + st^2, t^3, 0)^t$$

define subbundles $\mathcal{L}_0, \mathcal{L}_1 \subset V^* \otimes \mathcal{O}_B$, both isomorphic to $\mathcal{O}_B(-3)$. Let Y_0, Y_1 be their order-4 thickenings (4.4). They are finite flat of rank 712 over B and Zariski locally trivial with the same Artin algebra as fibre. Their nilradical graded vector bundles are isomorphic in every degree, although Y_0 and Y_1 are not isomorphic as abstract complex schemes.

Proof. Neither column vanishes, since its second entry is nonzero when $t \neq 0$, and its first is nonzero at $t = 0$. The left coefficient map to its image line $\mathbb{P}^1 \subset \mathbb{P}(V^*)$ is, in the coordinate $z = s/t$, respectively

$$f_0(z) = z^3, \quad f_1(z) = z^3 + z.$$

A local frame completing a generator of $\mathcal{L}_i$ to a basis of $V^* \otimes \mathcal{O}$ changes the coefficient line to a fixed line. This gives Zariski local product presentations of both thickenings. Alternatively, equality of the fibre isomorphism classes follows from transitivity of $\text{GL}(V)$ on nonzero coefficient lines. The rank formula is $\binom{13}{4} - 3 = 712$.

Let $E = V^* \otimes \mathbb{C}^3$, of dimension nine. For $j < 4$, the degree- j nilradical quotient is $\text{Sym}^j E \otimes \mathcal{O}_B$. Multiplication by the fixed determinant embeds E as a fixed nine-dimensional subspace of $\text{Sym}^4 E$.

Choosing a constant linear complement gives, for the top quotient,

$$\mathfrak{n}_i^4 \simeq \mathcal{O}_B^{486} \oplus ((V^* \otimes \mathcal{O}_B)/\mathcal{L}_i)^{\oplus 3}.$$

Each column has only its first two entries nonzero, and these two entries have no common zero. The rank-one quotient of $\mathcal{O}_B^2$ by $\mathcal{O}_B(-3)$ is consequently $\mathcal{O}_B(3)$, by determinants. Thus the displayed top bundle is $\mathcal{O}_B^{489} \oplus \mathcal{O}_B(3)^{\oplus 3}$ for both i . The assertions about every graded vector bundle follow. They are not assertions that the graded *algebra* structures agree.

If an abstract isomorphism $Y_0 \simeq Y_1$ existed, Theorem 4.3 would identify their coefficient maps by one automorphism of B and one constant projective transformation of V^* . Both maps have as image the same projective line, so the latter transformation would restrict to a projective automorphism of that line. Hence f_0 and f_1 would be equivalent by independent projective changes in source and target. This is impossible. The map f_0 has two ramification points, each of index three. The map f_1 has the two simple zeros of $3z^2 + 1$, each of index two, and infinity, of index three. Ramification indices are invariant under those projective changes. Thus the unmarked algebra scheme distinguishes these two globally different coefficient maps even though its fibre algebras, conormal bundle, abstract first-relation bundle, and all graded vector bundles agree. $\square$

4.3. Pencils and the category of their automorphisms. For pencils, take $q = 2p$, $p = N - 2$, and $F(H) = \bigwedge^{N-2} \text{Sym}^2 H$. By Lemma 6.2 this is a single Schur module of dimension $M = \binom{N}{2}$, and Lemma 9.1 identifies its coefficient subbundle with $\mathcal{L}_{\mathcal{R}} = \bigwedge^p (\text{Sym}^2 V/\mathcal{R})^*$. Its rank is one, so the strict-support hypothesis holds for every pencil. Exterior duality recovers the subbundle $\mathcal{R}$ from that line. Theorem 4.3 therefore proves the unrestricted moving-pencil inverse, including its actual source-bundle conclusion. This is a simultaneous reconstruction theorem for a map into a coefficient Grassmannian, not just a classification of its individual fibres.

Corollary 4.5 (Algebraic automorphism sequence for a pencil). *Let $\mathfrak{A}_R^{[d]}$ be the unmarked local algebra of Theorem 10.3, and let $G_R = \{g \in \text{GL}(V) : (\text{Sym}^2 g)R = R\}$. There is a split exact sequence of complex affine algebraic groups*

$$1 \longrightarrow U_R \longrightarrow \text{Aut}(\mathfrak{A}_R^{[d]}) \longrightarrow (G_R \times \text{GL}(U))/\mathbb{G}_m \longrightarrow 1, \quad (4.6)$$

where $\mathbb{G}_m$ is the diagonal scalar subgroup. Its intrinsic quotient is the closed stabilizer of K_R in the general linear group of the cotangent space. In that form the sequence represents automorphisms after extension to every commutative complex algebra. Its splitting is homogeneous after choosing generators, and U_R has the intrinsic description and dimension in Proposition 2.2.

Proof. Proposition 2.2 gives the represented sequence with quotient H_{K_R} . The coefficient inverse and Lemma 4.1 identify its complex points with the image of the left-right action; transposition is excluded by the support ranks $(1, M)$. Use the contragredient of the normal action to obtain a homomorphism on the cotangent space. Its kernel is exactly the diagonal scalar subgroup: a tensor product acting identically has both factors scalar, with reciprocal actions on the dual factors. The image is a closed algebraic subgroup and all the groups in question are smooth in characteristic zero [10, Lemma 39.8.2]. Thus the stabilizer and this image agree as closed subgroup schemes, not only on complex points. The induced quotient identification is an isomorphism of algebraic groups: after division by the kernel it is a bijective separable homomorphism between smooth groups. The splitting and kernel are those already constructed by substitutions. $\square$

For a positive-dimensional projective reduction, Theorem 9.4 is used as an exact sequence of groups of complex scheme automorphisms, with its stated filtered kernel. We do not identify that sequence with a finite-type affine group scheme or a moduli-stack equivalence. Likewise, the marked construction over nonreduced parameter bases later in the paper retains its specified

relative socle: it does not replace that ideal by the absolute nilradical. These distinctions preserve the full unmarked geometric assertion while specifying its category.

5. AN INTRINSIC COEFFICIENT PRINCIPLE

Extracting coefficient data from an abstract scheme and reconstructing a defining relation space from those data are distinct operations. The criterion below isolates the first. Its hypotheses will be verified directly for the multiplication schemes and then on curves.

5.1. The structural criterion. Let X be a finite-type complex scheme with a closed subvariety B specified by a construction preserved by every abstract scheme isomorphism. Write $\mathfrak{b} \subset \mathcal{O}_X$ for its ideal and

$$\mathcal{G} = \bigoplus_{r \geq 0} \mathfrak{b}^r / \mathfrak{b}^{r+1}, \quad \mathcal{A} = \text{Sym}_{\mathcal{O}_B}(\mathcal{G}_1), \quad \mathcal{I} = \ker(\mathcal{A} \twoheadrightarrow \mathcal{G}).$$

Assume the following conditions, for fixed integers $n \geq 2, m \geq 1$.

- (i) B is connected and projective, $H^0(B, \mathcal{O}_B) = \mathbb{C}$, and $\mathcal{G}_1$ is locally free. Its dual admits a presentation $V \otimes \mathcal{U}^*$, where $\dim V = \text{rank } \mathcal{U} = n$.
- (ii) The reduction of $\text{Spec}_B \mathcal{G}$ in its degree-one ambient bundle is the generic determinant cone. The bundle $\mathbb{P}_B(\mathcal{U}^*)$ is not projectively trivial. Thus precisely one of the two ruling parameter bundles of its rank-one locus is projectively trivial.
- (iii) If $\mathcal{D} \subset \mathcal{A}$ is the ideal of that reduced cone and $\mathcal{L} = \mathcal{D}_n$, then $\mathcal{I} = \mathcal{D}\mathcal{J}$, where $\mathcal{J}$ is generated in degree m . Under the Cauchy decomposition in any local frame of $\mathcal{U}$,

$$\mathcal{J}_m = \bigoplus_{\lambda \in \Lambda} L_\lambda \otimes \mathbb{S}_\lambda \mathcal{U} \subset \text{Sym}^m(V^* \otimes \mathcal{U}), \quad L_\lambda \subset \mathbb{S}_\lambda V^*, \quad (5.1)$$

where Λ is a specified collection of partitions of m of length at most n , and the left subspaces are constant over B . Zero subspaces are permitted. The Cauchy inclusions, not just their representation labels, are the coefficient realizations $\beta \otimes \xi \mapsto [T \mapsto \beta(\mathbb{S}_\lambda(T)\xi)]$.

These conditions are hypotheses on the normal cone and its residual coefficient module. They do not assume that an abstract isomorphism extends to an ambient space, preserves a chosen frame, or already acts on all the left subspaces through a common projective transformation.

Theorem 5.1 (Intrinsic coefficient reconstruction). *Suppose X, B satisfy (i)–(iii). The abstract scheme X determines the collection $(L_\lambda)_{\lambda \in \Lambda}$ up to one common projective coordinate transformation on V . More precisely, an isomorphism between two such schemes induces a constant projective isomorphism $[g] : \mathbb{P}(V) \rightarrow \mathbb{P}(V')$ such that*

$$L_\lambda = \mathbb{S}_\lambda(g)^* L'_\lambda \quad (\lambda \in \Lambda).$$

The equality means equality of subspaces; scalar changes of a linear lift of $[g]$ do not affect it.

Proof. An isomorphism transports B , every power of $\mathfrak{b}$, $\mathcal{G}_1$, and the surjection $\mathcal{A} \twoheadrightarrow \mathcal{G}$. In particular it transports $\mathcal{I}$ in the intrinsic degree-one ambient bundle. Iterated reduced singular loci of the determinant cone give its rank-one locus. Lemma 3.1 reconstructs its two ruling parameter schemes, including their scheme structure. Condition (ii) distinguishes the trivial ruling intrinsically. Lemma 3.2 then gives one constant projective map $[g]$. After choosing a constant linear lift and local frames on both bases, the ambient bundle map has the form $T \mapsto gTh^{-1}$. These local choices are made only after constructing the graded algebra, its degree-one part, and the ruling.

Locally $\mathcal{D} = (d)$ with $d = \det T$. Its leading coefficient is a unit, so it is a nonzerodivisor over every coefficient ring. Hence $(\mathcal{I} : \mathcal{D}) = \mathcal{J}$, and

$$\mathcal{J}_m \simeq \mathcal{L}^* \otimes \mathcal{I}_{n+m}.$$

This recovers the residual module without choosing a scalar determinant equation. Under $T \mapsto gTh^{-1}$, the Cauchy coefficient map transports a tensor by $\mathbb{S}_\lambda(g)^* \otimes \mathbb{S}_\lambda(h^{-1})$. Projection to the specified Cauchy summand, followed by contraction against its whole right dual, therefore transports exactly L'_λ to L_λ . The right map is invertible and has no effect on that left image. Both determinant-line transition functions and changes of a lift of $[g]$ act by scalar units. Since every partition has size m , this ambiguity is common in degree and disappears on subspaces. This proves the assertion. $\square$

6. QUADRATIC RELATION SPACES AND INTRINSIC RECONSTRUCTION OF PENCILS

The multiplication construction itself supplies a wider natural class for the coefficient criterion. In this section the number of quadratic relations is allowed to vary independently of the embedding dimension. For two relations the resulting theorem recovers every pencil of quadrics, including singular pencils. No extra block is added to the multiplication map.

6.1. Automatic coefficient extraction for quadratic multiplication. Let $\dim V = n \geq 3$, $W = \text{Sym}^2 V$, $N = n(n+1)/2$, and $R \in \text{Gr}(r, W)$, where $1 \leq r < N - n$. Put $p = N - r > n$, $S = W/R$, and

$$A_R = \mathbb{C} \oplus V \oplus S, \quad X_R = \text{Gr}(n, V \oplus S).$$

The only nonzero product in the maximal ideal is the quotient map $\gamma: \text{Sym}^2 V \rightarrow S$. Write $\mathcal{K}$ for the tautological bundle on X_R , and use the same definition (1.1) of $\widehat{D}_R$, with A_R in place of B_R . Thus $\widehat{D}_R$ is the failure scheme of generating A_R by a varying n -plane and the unit. The notation in this section agrees with the web construction when $(n, r) = (4, 4)$.

Let $\Lambda_{n,p}$ denote the partitions occurring in the classical multiplicity-free decomposition

$$F(V) := \bigwedge^p \text{Sym}^2 V = \bigoplus_{\lambda \in \Lambda_{n,p}} \mathbb{S}_\lambda V. \quad (6.1)$$

Every partition has size $2p$; terms of length greater than n are omitted. Choose a nonzero volume form on S and let $\beta_R \in F(V)^*$ be its pullback by $\bigwedge^p \gamma$. Its component in $\mathbb{S}_\lambda V^*$ is denoted by $\beta_{R,\lambda}$. Only the lines of the nonzero components will be used, so the volume choice is immaterial.

Theorem 6.1 (Automatic intrinsic coefficient extraction). *For every $R \in \text{Gr}(r, \text{Sym}^2 V)$ in the stated range, the abstract scheme $\widehat{D}_R$ determines*

$$(\mathbb{C}\beta_{R,\lambda})_{\lambda \in \Lambda_{n,p}}$$

up to one common projective transformation of V , with zero components recorded as zero subspaces. More precisely, the intrinsic construction recovers the base $B = \text{Gr}(n, S)$, its degree-one normal-cone bundle, and the residual ideal generated in degree $2p$ with coefficient module

$$\mathcal{J}_{2p} = \bigoplus_{\lambda \in \Lambda_{n,p}} \mathbb{C}\beta_{R,\lambda} \otimes \mathbb{S}_\lambda \mathcal{U} \subset \text{Sym}^{2p}(V^* \otimes \mathcal{U}), \quad (6.2)$$

where $\mathcal{U}$ is the tautological rank- n bundle on B . Consequently all geometric hypotheses of Theorem 5.1 hold for this entire Grassmannian family, not just for smooth webs.

Proof. On a frame chart of $\mathcal{K}$, write its inclusion as $\begin{pmatrix} M \\ A_0 \end{pmatrix}$, with M an $n \times n$ matrix. Since A_R is cube-zero, columns of degree greater than two vanish. After retaining the scalar, linear, and quadratic columns, multiplication has the matrix

$$\begin{pmatrix} 1 & 0 & 0 \\ 0 & M & 0 \\ 0 & A_0 & \gamma \text{Sym}^2 M \end{pmatrix}. \quad (6.3)$$

A nonzero maximal minor must use the scalar column and all n linear columns: these are the only columns meeting the first $1 + n$ rows. Expanding along those rows gives the identity of ideals

$$I_{\widehat{D}_R} = (\det M)I_p(\gamma \operatorname{Sym}^2 M). \quad (6.4)$$

The same expansion works for every $m \geq 2$. Where M is invertible, the quadratic map is surjective. Hence the reduction of $\widehat{D}_R$ is exactly the Schubert divisor where M is singular, independent of R after identifying the vector spaces S .

Successive reduced singular loci of that divisor are the projection rank loci. Their last nonempty member is $B = \operatorname{Gr}(n, S)$. For clarity, the usual determinantal assertion here is local: at a projection-rank- k point, an invertible k -minor splits away, leaving a generic square $(n - k)$ -matrix and smooth coordinates. Every remaining entry can vary independently by moving the kernel of the projection in V modulo its image. Thus its reduced singular locus is the next rank locus. Since $p > n$, the final base is nonempty, connected and projective, with $H^0(B, \mathcal{O}_B) = \mathbb{C}$.

Near B , a Grassmannian chart consists of a varying plane in S and its graph $T : \mathcal{U} \rightarrow V$. Formula (6.4) is independent of the other graph coordinates and homogeneous in T . It therefore gives its actual normal-cone ideal, not merely an initial reduced support:

$$C_B \widehat{D}_R \subset \operatorname{Tot} \operatorname{Hom}(\mathcal{U}, V), \quad \mathcal{I} = (\det T)\mathcal{J}, \quad \mathcal{J} = I_p(\gamma \operatorname{Sym}^2 T). \quad (6.5)$$

There are no degree-one equations. The ambient bundle in (6.5) is consequently the dual of the intrinsically recovered degree-one conormal. Its reduced ideal is the determinant ideal $\mathcal{D}$. The determinant polynomial is monic for a suitable monomial order, so cancellation gives $(\mathcal{I} : \mathcal{D}) = \mathcal{J}$. This is an intrinsic ideal quotient; choosing a generator of the determinant line is not part of the recovered datum.

The rank-one locus of the reduced cone determines the two ruling parameter schemes by Lemma 3.1. They are $\mathbb{P}(V) \times B$ and $\mathbb{P}_B(\mathcal{U}^*)$. Exactly the first is projectively trivial. Indeed a Schubert line in $\operatorname{Gr}(n, S)$ restricts $\mathcal{U}^*$ to $\mathcal{O}(1) \oplus \mathcal{O}^{n-1}$. A trivial projective bundle on that line would make this bundle a line-bundle twist of $\mathcal{O}^n$; its determinant would then have degree divisible by n , rather than degree one. Thus the trivial ruling is intrinsically oriented. Lemma 3.2 applies: an abstract isomorphism gives a morphism $B \rightarrow \operatorname{PGL}(V)$, constant because B is connected projective and $\operatorname{PGL}(V)$ is affine. Local linear lifts and right changes of frame give $T \mapsto gTh^{-1}$ for this one fixed $[g]$.

It remains to prove (6.2), rather than assume it as a condition on the example. The minors are the image, for fixed β_R , of the universal coefficient map

$$F(V)^* \otimes F(U) \longrightarrow \operatorname{Sym}^{2p}(V^* \otimes U), \quad \beta \otimes \xi \longmapsto [T \mapsto \beta(F(T)\xi)]. \quad (6.6)$$

In characteristic zero, the polynomial functor F has the multiplicity-free decomposition (6.1); see [3, Theorem 3.5]. The Cauchy decomposition of the target contains each diagonal $\mathbb{S}_\lambda V^* \otimes \mathbb{S}_\lambda U$ once, and no off-diagonal pair. Equivariance kills every off-diagonal pair. On each diagonal pair the coefficient map is nonzero: identify U with V , evaluate at $T = 1$, and choose β, ξ in that summand with $\beta(\xi) \neq 0$. Schur's lemma then identifies its image with the specified Cauchy summand. This is the actual matrix-coefficient inclusion, not merely a dimension count. Keeping β_R fixed proves (6.2). All generators have degree $2p$, and at least one minor is nonzero since γ is surjective.

Under $T \mapsto gTh^{-1}$, the right map acts within the full factor $\mathbb{S}_\lambda U$. Contracting against its dual therefore leaves exactly the left coefficient line transformed by $\mathbb{S}_\lambda(g)^*$. Determinant-line scalars and local lift ambiguities do not change these lines. This proves the theorem. $\square$

The theorem does not assert that separately projectivized Schur components always determine their relative scalars. For webs this is the recombination problem solved in the separate research archive. For pencils that issue disappears altogether.

6.2. Every pencil, without a regularity hypothesis.

Lemma 6.2 (The exterior square of the quadratic representation). *For $n \geq 2$, $\bigwedge^2 \text{Sym}^2 V = \mathbb{S}_{(3,1)} V$ is irreducible. Consequently $\bigwedge^{N-2} \text{Sym}^2 V$ is irreducible as well.*

Proof. The vector $x_1^2 \wedge x_1 x_2$ is a highest-weight vector of weight $(3, 1)$. By complete reducibility it generates a copy of $\mathbb{S}_{(3,1)} V$. The hook formula gives

$$\dim \mathbb{S}_{(3,1)} V = \frac{n(n-1)(n+1)(n+2)}{8} = \binom{N}{2}.$$

This exhausts the exterior square. The last assertion follows by exterior duality. These are classical representation identities. $\square$

Theorem 6.3 (Intrinsic reconstruction of all quadratic pencils). *Let $n \geq 3$ and let R, R' be two-dimensional subspaces of $\text{Sym}^2 V$. For the natural multiplication-failure schemes of their cube-zero algebras of Hilbert function $(1, n, N-2)$,*

$$\widehat{D}_R \simeq \widehat{D}_{R'} \iff R' = gR \text{ for some } g \in \text{PGL}(V). \quad (6.7)$$

No pencil, base locus, or discriminant is required to be smooth or regular. The isomorphism on the left is abstract and carries no supplied ambient embedding, base, or tensor-factor marking. All the reduced schemes are isomorphic to the same Schubert divisor.

Proof. Here $p = N-2 > n$. By Theorem 6.1 and Lemma 6.2, the normal cone has only one left coefficient line to recover: the entire line $\mathbb{C}\beta_R$. The canonical duality

$$\bigwedge^{N-2} W^* \simeq \bigwedge^2 W \otimes (\det W)^* = \bigwedge^2 W \otimes (\det V)^{-(n+1)} \quad (6.8)$$

sends that line to the Pluecker line of R , with the same projective transformation g . Indeed the pullback of a quotient volume is the annihilator-volume dual of the kernel R . The common determinant character is a scalar on projective space. A nonzero decomposable two-vector recovers its plane as $\{w \in W : w \wedge w_R = 0\}$. Thus an abstract scheme isomorphism implies $R' = gR$.

Conversely such a linear coordinate change induces the quotient isomorphism $S_R \rightarrow S_{R'}$, the algebra isomorphism $A_R \rightarrow A_{R'}$, and the corresponding Grassmannian isomorphism. It intertwines multiplication and hence preserves its zeroth Fitting ideal. This proves the reverse implication. The assertion about reductions was proved in (6.4). $\square$

6.3. Discriminants, elementary divisors, and hyperelliptic moduli. A regular pencil means one containing an invertible symmetric matrix; it need not have a reduced discriminant. The classical invariant of such a pencil consists of its discriminant points together with their Jordan-block partitions, usually called its Weierstrass–Segre data. We use the root-labelled partitions, not a pairing of an unlabelled list with an independently unlabelled divisor. Their classification is classical; compare [4, Theorem 1.1, Corollary 2.1 and §2]. The following consequence identifies precisely what the intrinsic failure scheme remembers beyond the discriminant.

Corollary 6.4 (Intrinsic recovery of elementary divisors). *For a regular pencil in any dimension $n \geq 3$, the abstract scheme $\widehat{D}_R$ determines all root-labelled elementary divisors of a matrix pencil, up to projective reparametrization of its parameter line. Conversely these data determine its abstract failure scheme. In particular the invariant distinguishes different Jordan partitions over a fixed multiple discriminant root.*

Proof. The first implication follows by reconstructing R , then choosing an invertible member A and a complementary member B . For completeness, the relevant classical sufficiency can be seen

directly. Put $C = A^{-1}B$; it is self-adjoint for A . If $hC = C'h$ identifies the Jordan forms for another pair (A', B') , set

$$H = A^{-1}h^t A' h.$$

Then H is invertible, A -self-adjoint, and commutes with C . There is a polynomial k in H with $k^2 = H$: on each nonzero eigenvalue choose a square root and truncate its Taylor series to the size of the largest Jordan block, then combine by the Chinese remainder theorem. Thus k is invertible, self-adjoint, and commutes with C . The map hk^{-1} is an isometry from A to A' and intertwines C, C' , so it simultaneously identifies A, B with A', B' .

Changing the basis of the pencil replaces C by a fractional linear expression with invertible denominator. At each eigenvalue its derivative is nonzero; hence each Jordan-block size is preserved while its root moves by the same projective transformation. A parameter change can avoid infinity for both finite spectra. Root-labelled elementary divisors therefore suffice, and (6.7) proves the converse for the failure schemes. This argument recalls the classical classification; the new input is recovery from an unmarked abstract failure scheme. $\square$

Example 6.5 (Identical discriminant, distinct intrinsic failure schemes). Choose distinct nonzero $\mu_3, \dots, \mu_n$, and put

$$S_2 = \begin{pmatrix} 0 & 1 \\ 1 & 0 \end{pmatrix}, \quad J_2 = \begin{pmatrix} 0 & 1 \\ 0 & 0 \end{pmatrix}.$$

Consider the symmetric pairs

$$\begin{aligned} A_s &= I_n, & B_s &= \text{diag}(0, 0, \mu_3, \dots, \mu_n), \\ A_j &= S_2 \oplus I_{n-2}, & B_j &= S_2 J_2 \oplus \text{diag}(\mu_3, \dots, \mu_n). \end{aligned}$$

Their determinant binary forms agree up to a nonzero scalar:

$$\det(sA_s + tB_s) = -\det(sA_j + tB_j) = s^2 \prod_{i=3}^n (s + \mu_i t). \quad (6.9)$$

At the unique double root the ranks are respectively $n-2$ and $n-1$, so no simultaneous projective equivalence exists. The two root partitions are $(1, 1)$ and (2) . Their abstract failure schemes are therefore nonisomorphic although both their reduced failure schemes and their discriminant divisors, including multiplicities, agree. This realizes inside the same multiplication construction the classical loss of elementary-divisor information under passage to the determinant.

Corollary 6.6 (Hyperelliptic isomorphism classes). *Fix $g \geq 2$ and $n = 2g + 2$. On the locus of pencils with simple discriminant, $\widehat{D}_R$ gives an injective invariant of the isomorphism classes of smooth hyperelliptic curves of genus g . All these failure schemes have the same reduction; their nonreduced structures distinguish a $(2g - 1)$ -dimensional family of curve classes. The pencil invariant also distinguishes the two regular boundary pencils in Example 6.5, where their singular discriminant double covers alone do not.*

Proof. The classical curve associated with a simple-discriminant pencil is the double cover of its parameter line branched at $\det(sA + tB) = 0$; see [6, §4.2, equation (4.3)]. Choose an invertible member. Distinct eigenvalues of $A^{-1}B$ have mutually A -orthogonal eigenlines, each nonisotropic, so simultaneous diagonalization identifies the pencil with $\langle \sum x_i^2, \sum \lambda_i x_i^2 \rangle$. The unordered branch divisor modulo PGL_2 therefore determines the pencil, and every reduced divisor of degree $2g + 2$ arises. A smooth curve of genus at least two has a unique degree-two map to a rational curve up to automorphism of that curve: two distinct such maps would give a birational map to a curve of bidegree $(2, 2)$ in $\mathbb{P}^1 \times \mathbb{P}^1$, whose arithmetic genus is one. Thus curve isomorphism is exactly branch-divisor equivalence. Applying Theorem 6.3 gives the asserted injectivity and well-definedness.

Ordered distinct branch points have dimension $2g + 2$. Normalizing three of them removes the three-dimensional projective action; taking the finite permutation quotient leaves dimension $2g - 1$. Equality of all reductions follows from (6.4). Finally the two binary forms in (6.9) differ only by a scalar, a square over $\mathbb{C}$, and define isomorphic double covers, whereas their failure schemes have already been separated. $\square$

These are statements about complex isomorphism classes. They do not identify moduli stacks, assert a universal family over a coarse moduli space, or eliminate the different automorphism groups of pencils and curves. Nor is the classical pencil–hyperelliptic correspondence a new Torelli theorem. The conclusion here is reconstruction of those objects and of the elementary-divisor information on their boundary from the abstract nonreduced multiplication-failure scheme, while all reduced failure data are fixed.

7. THE EXACT INFINITESIMAL ORDER OF PENCIL RECONSTRUCTION

The all-pencil theorem admits a finite-order strengthening. It identifies the first infinitesimal neighbourhood that retains the pencil, and shows that every preceding neighbourhood is independent of it. We use ideal-adic neighbourhoods of the deepest stratum; these are not arc spaces or jet schemes.

Throughout this section let $n \geq 3$, $W = \text{Sym}^2 V$, and

$$N = \binom{n+1}{2}, \quad p = N - 2, \quad d = n + 2p = n^2 + 2n - 4. \quad (7.1)$$

For a pencil $R \subset W$, put $S_R = W/R$ and $B = \text{Gr}(n, S_R) \subset \widehat{D}_R$. This notation for the socle Grassmannian is local to this section; the cube-zero algebra is denoted by A_R , as in Section 6. Write $\mathfrak{b} \subset \mathcal{O}_{\widehat{D}_R}$ for the ideal of B , and define

$$Z_R^{[k]} = (B, \mathcal{O}_{\widehat{D}_R}/\mathfrak{b}^{k+1}) \quad (k \geq 0). \quad (7.2)$$

Thus $Z_R^{[0]} = B$. No embedding or identification of its reduction is supplied with the abstract scheme $Z_R^{[k]}$.

Theorem 7.1 (Sharp finite-neighbourhood reconstruction). *For any two pencils $R, R' \subset \text{Sym}^2 V$, the following conditions are equivalent:*

$$\begin{aligned} Z_R^{[d]} \simeq Z_{R'}^{[d]} &\iff \widehat{D}_R \simeq \widehat{D}_{R'} \\ &\iff R' = gR \text{ for some } g \in \text{PGL}(V). \end{aligned} \quad (7.3)$$

In contrast, for every $0 \leq k < d$, all the schemes $Z_R^{[k]}$ are isomorphic. Consequently d is the smallest uniform order at which these neighbourhoods classify all pencils up to projective equivalence. Singular pencils are included.

7.1. The first relation in the conormal algebra. Let $\mathcal{U}$ be tautological on B . The open set of X_R where projection to S_R is injective is canonically $\text{Tot } E$, where $E = \text{Hom}(\mathcal{U}, V)$: a plane is the graph of a unique map from its image in S_R to V . In particular this is one open neighbourhood of all of B , not merely a collection of unrelated formal charts. Let $\mathfrak{a}$ be its zero-section ideal. The exact multiplication calculation (6.4) gives

$$\mathcal{I}_R = (\det T) I_p(\gamma_R \text{Sym}^2 T) \subset \text{Sym } E^*. \quad (7.4)$$

It is generated in degree d , with a nonzero degree- d piece K_R . Indeed each quadratic minor has degree $2p$, and an invertible T makes $\gamma_R \text{Sym}^2 T$ surjective. There are no terms of lower degree,

even on special pencils. Homogeneity is an equality of ideals in the actual graph neighbourhood, not just an equality of initial supports.

Lemma 7.2 (Intrinsic recovery from the first relation). *The abstract scheme $Z_R^{[d]}$ determines, as a graded $\mathcal{O}_B$ -algebra, the complete normal-cone algebra $\mathrm{gr}_{\mathfrak{b}} \mathcal{O}_{\widehat{D}_R}$.*

Proof. The reduction of $Z_R^{[d]}$ is B , since its zero-section ideal is nilpotent and B is reduced. Hence its nilradical $\mathfrak{n}$ is intrinsic. Put $L = \mathfrak{n}/\mathfrak{n}^2$. Formula (7.4) and $d > 1$ identify $L = E^*$. Multiplication in the given finite scheme recovers

$$K_R = \ker[\mathrm{Sym}^d L \longrightarrow \mathfrak{n}^d/\mathfrak{n}^{d+1}]. \quad (7.5)$$

The last denominator is zero, but retaining it displays the grading. No splitting of the reduction map is needed to form this kernel. Because $\mathcal{I}_R$ is generated in this single degree,

$$\mathrm{gr}_{\mathfrak{b}} \mathcal{O}_{\widehat{D}_R} = \mathrm{Sym}_{\mathcal{O}_B} L/(K_R). \quad (7.6)$$

Here (K_R) denotes the homogeneous ideal generated by K_R , not its radical. The right side is intrinsic to the finite scheme, proving the assertion. $\square$

Proof of Theorem 7.1. For $k < d$, equations (7.4) and (7.2) give

$$\mathcal{O}_{Z_R^{[k]}} = \mathrm{Sym} E^*/\mathfrak{a}^{k+1}.$$

A linear identification $S_R \simeq S_{R'}$ identifies B , its tautological bundle, and E . It therefore identifies every one of these lower-order neighbourhoods.

An abstract isomorphism at order d induces an isomorphism of reductions, of conormal bundles, and, by Lemma 7.2, of complete graded normal cones. We spell out why the reconstruction in Theorem 6.1 requires no information outside this cone. Its reduced support is the square-matrix determinant cone; its iterated singular loci give the rank-one cone. The relative scheme of maximal linear spaces in that cone has the two rulings $\mathbb{P}(V) \times B$ and $\mathbb{P}_B(\mathcal{U}^*)$. Exactly the first is projectively trivial, as is seen by restriction to a Schubert line. Thus it is intrinsically oriented. The induced projective transformation of the trivial factor is constant on the connected projective base. Local lifts have the form $T \mapsto gTh^{-1}$ for this one $[g]$. The determinant ideal line is intrinsic, so its ideal quotient in (7.4) recovers the residual ideal. Finally Lemma 6.2 identifies the entire left coefficient line $\mathbb{C}\beta_R$, and exterior duality (6.8) recovers the Pluecker line of R . These are operations on the recovered normal cone; we do not assume that its isomorphism extends to X_R or $\widehat{D}_R$. They imply $R' = gR$.

Conversely such a projective transformation induces the algebra and failure-scheme isomorphisms, carrying the socle Grassmannian to the socle Grassmannian and preserving all its ideal powers. This proves (7.3), also using Theorem 6.3. For sharpness, there are inequivalent pencils in every dimension under consideration: $\langle x_1^2, x_1x_2 \rangle$ has a common linear factor, whereas $\langle x_1^2, x_2^2 \rangle$ does not. Their neighbourhoods are nevertheless isomorphic at every order below d . Thus no smaller order classifies all pencils. $\square$

7.2. A closed immersion of the full parameter scheme. The finite-order result has a scheme-theoretic parameter version, which records infinitesimal variation as well as closed points. Here tensor coordinates are fixed. This additional qualification is important: we are not asserting an equivalence of unmarked moduli stacks.

Fix an auxiliary n -dimensional vector space U , and put

$$F(V) = \bigwedge^p \mathrm{Sym}^2 V, \quad F(U) = \bigwedge^p \mathrm{Sym}^2 U, \quad M = \dim F(U) = \binom{N}{2}, \quad H = \mathrm{Hom}(U, V).$$

Choose a nonzero scalar representative δ of the determinant polynomial on H ; its scalar has no effect on the subspaces below. Let $K_R \subset \text{Sym}^d H^*$ be the degree- d space of $(\det T)I_p(\gamma_R \text{Sym}^2 T)$ in these fixed coordinates.

Theorem 7.3 (Faithful finite-order parameterization). *The assignment*

$$\Phi : \text{Gr}(2, \text{Sym}^2 V) \longrightarrow \text{Gr}(M, \text{Sym}^d H^*), \quad R \longmapsto K_R, \quad (7.7)$$

is a closed immersion. It is defined on the whole Grassmannian, including singular pencils. Its construction and its inverse on the image commute with arbitrary base change over $\mathbb{C}$. In particular it separates families over nonreduced bases and has injective differential at every pencil.

Lemma 7.4 (Tensoring a line with a fixed space). *For finite-dimensional vector spaces A, B_0 , with $b = \dim B_0 > 0$, the map $\mathbb{P}(A) \rightarrow \text{Gr}(b, A \otimes B_0)$, $\ell \mapsto \ell \otimes B_0$, is a closed immersion.*

Proof. Choose bases $a_0, \dots, a_t$ and a basis of B_0 . Above the projective chart $\ell = \mathbb{C}(a_0 + \sum x_i a_i)$, the subspace is the graph of the block matrices $x_i I_b$. The corresponding Grassmannian chart has arbitrary $b \times b$ blocks. The image is cut out by their off-diagonal entries and the differences between their diagonal entries; an inverse reads one diagonal entry in each block. These are scheme-theoretic equations over any $\mathbb{C}$ -algebra, not just pointwise conditions. The preimage of this Grassmannian chart is exactly the stated projective chart, since the projection has determinant a_0^b . Such charts cover the image. The map is proper because its source is projective, so these chartwise closed immersions give a global closed immersion. $\square$

Proof of Theorem 7.3. The coefficient map (6.6) becomes

$$\mu : F(V)^* \otimes F(U) \longrightarrow \text{Sym}^{2p} H^*.$$

It is injective: by Lemma 6.2 its source is irreducible for $\text{GL}(V) \times \text{GL}(U)$, and evaluation at an isomorphism $U \simeq V$ shows that the equivariant map is nonzero. Multiplication by the nonzero polynomial δ is injective. Thus $\nu = \delta\mu$ is a fixed linear inclusion and

$$K_R = \nu(\mathbb{C}\beta_R \otimes F(U)). \quad (7.8)$$

In particular $\dim K_R = M$, also on every singular pencil. By exterior duality the quotient-volume map $R \mapsto \mathbb{C}\beta_R$ is the Pluecker closed immersion $\text{Gr}(2, W) \hookrightarrow \mathbb{P}(F(V)^*)$. Factor Φ through this map, the closed immersion of Lemma 7.4, and the Grassmannian closed immersion induced by ν . This proves the theorem. All are morphisms of schemes defined by the displayed polynomial and linear equations. A closed immersion is a monomorphism and remains one after arbitrary base change. This proves the family assertion, not merely separation of complex-valued points. $\square$

Corollary 7.5 (A flat family of finite normal algebras). *Over $\text{Gr}(2, W)$, the algebras*

$$Q_R = \text{Sym} H^* / ((K_R) + \mathfrak{m}^{d+1}), \quad \mathfrak{m} = \bigoplus_{j>0} \text{Sym}^j H^*, \quad (7.9)$$

form a finite locally free family of rank

$$\binom{n^2 + d}{d} - \binom{N}{2}.$$

In fixed coordinates this family determines every base-changed family of pencils. All its truncations below degree d are constant families.

Proof. Theorem 7.3 makes the spaces K_R a rank- M subbundle of the constant degree- d bundle, with locally free quotient. In degree $j < d$ the algebra is $\text{Sym}^j H^*$, and its degree- d piece is $\text{Sym}^d H^* / K_R$; higher degrees vanish. This proves local freeness and base-change compatibility. Summing the ranks gives the displayed binomial identity. The kernel of the coordinate quotient

in degree d recovers K_R , and Φ recovers the pencil family. This is a statement about coordinate quotient algebras; it does not discard their coordinate maps and then infer a family reconstruction theorem. $\square$

7.3. Classical moduli at a sharp order.

Corollary 7.6 (Hyperelliptic classes in finite neighbourhoods). *Let $g \geq 2$, $n = 2g + 2$, and $d = 4g^2 + 12g + 4$. For pencils with simple discriminant, $Z_R^{[d]}$ and $Z_{R'}^{[d]}$ are isomorphic if and only if their associated hyperelliptic curves are isomorphic. Every lower-order neighbourhood is independent of the curve. Thus the same $(2g - 1)$ -dimensional family of curve classes is invisible to all these lower-order neighbourhoods and is separated at order d .*

Proof. Combine Theorem 7.1 with Corollary 6.6, and substitute $n = 2g + 2$ in (7.1). $\square$

The order here concerns a neighbourhood of the entire projective socle Grassmannian. Its global ruling distinguishes the constant V -factor from the tautological factor. A single unmarked Artinian normal slice does not supply this orientation; the coordinate statement of Corollary 7.5 is deliberately different. The result is therefore not a claim that one unmarked local algebra replaces the projective family. The matrix-pencil classification and the hyperelliptic correspondence remain classical. The assertion proved here is that this classical information first appears at the specified order in the intrinsic multiplication-failure neighbourhoods.

8. RECONSTRUCTION ON A PROJECTIVE CURVE

The global part of the inverse argument has a more economical geometric realization. It uses a proper base to make the projective left frame constant, and a nontrivial right projective bundle to distinguish the rulings. Neither step requires the whole socle Grassmannian. We now replace that Grassmannian by a curve. In particular, a neighbourhood supported on one Schubert line already retains every pencil.

Let C be a smooth connected projective complex curve, and let $\mathcal{U}$ be a rank- n bundle on C , where $n \geq 3$. Assume that $\mathbb{P}_C(\mathcal{U}^*)$ is not projectively trivial. A sufficient numerical condition is

$$\deg \det \mathcal{U} \not\equiv 0 \pmod{n}. \quad (8.1)$$

Indeed a projectively trivial rank- n bundle is a line-bundle twist of a trivial bundle, so its determinant degree is divisible by n . Fix V of dimension n , put $E = \text{Hom}(\mathcal{U}, V)$, and let $\mathfrak{a}$ be the zero-section ideal in $\text{Sym } E^*$. For a pencil $R \subset \text{Sym}^2 V$, the quotient $\gamma_R : \text{Sym}^2 V \rightarrow S_R$ defines the homogeneous ideal

$$\mathcal{I}_{R,C} = (\det T) I_p(\gamma_R \text{Sym}^2 T), \quad p = \binom{n+1}{2} - 2, \quad (8.2)$$

where $T : \mathcal{U} \rightarrow V$ is the universal map on $\text{Tot } E$. The expression is independent of local frames: both determinant multiplication and the maximal-minor construction are bundle constructions. Define

$$Y_{R,C}^{[k]} = \text{Spec}_C(\text{Sym } E^* / (\mathcal{I}_{R,C} + \mathfrak{a}^{k+1})).$$

Only its abstract complex scheme is retained, not its projection to C , its embedding in $\text{Tot } E$, or either tensor factor.

Theorem 8.1 (Curve-supported reconstruction). *For fixed $(C, \mathcal{U}, V)$ as above and $d = n^2 + 2n - 4$, every pair of quadratic pencils satisfies*

$$Y_{R,C}^{[d]} \simeq Y_{R',C}^{[d]} \iff R' = gR \text{ for some } g \in \text{PGL}(V). \quad (8.3)$$

All lower-order neighbourhoods are independent of R . The order- d schemes are finite locally free over C of rank $\binom{n^2+d}{d} - \binom{N}{2}$, where $N = \binom{n+1}{2}$. No regularity condition is imposed on the pencil.

Proof. On any frame chart, the ideal in (8.2) is generated in degree d . Its degree- d piece K_R is a subbundle of rank $\binom{N}{2}$, by the fixed coefficient inclusion (7.8); its quotient is locally free. The quotient algebra has its full symmetric powers in degrees less than d , has $\mathrm{Sym}^d E^*/K_R$ in degree d , and is zero above that degree. This proves the rank and the assertion about lower orders.

Its reduction is C , and its nilradical $\mathfrak{n}$ recovers $L = \mathfrak{n}/\mathfrak{n}^2$ and

$$K_R = \ker[\mathrm{Sym}^d L \longrightarrow \mathfrak{n}^d].$$

Thus the abstract finite-order scheme reconstructs the entire graded algebra $\mathrm{Sym} L/(K_R)$, by the same argument as Lemma 7.2. The ideal here means the homogeneous ideal generated by the recovered kernel, not its radical. No splitting of the reduction or supplied projection is used.

The reduction of this reconstructed cone is the determinant cone. Indeed its ideal has the determinant factor, and when T is invertible the quadratic quotient has full rank. Its rank-one Fano scheme has two ruling components, $\mathbb{P}(V) \times C$ and $\mathbb{P}_C(\mathcal{U}^*)$. The hypothesis distinguishes the first one intrinsically. An abstract isomorphism of neighbourhoods may induce a nonidentity automorphism of C ; pull back the second bundle along that automorphism. The oriented ruling map is nevertheless a morphism from C to PGL_n . It is constant, since C is connected projective and $H^0(C, \mathcal{O}_C) = \mathbb{C}$. Lemma 3.2 therefore supplies one constant projective left frame.

Cancel the intrinsic determinant ideal and apply the coefficient construction of Theorem 5.1. For pencils, Lemma 6.2 leaves just the single line $\mathbb{C}\beta_R$. Exterior duality (6.8) recovers R with respect to that same projective frame. This proves the forward implication in (8.3). Conversely a linear lift of g induces the bundle map on E and the quotient isomorphism on S_R , and carries all the displayed ideals and their truncations to the corresponding ideals for R' . This proves the converse. $\square$

Corollary 8.2 (One Schubert line suffices). *Let $\ell \simeq \mathbb{P}^1$ be any Schubert line in $B_R = \mathrm{Gr}(n, S_R)$, and use the canonical graph projection $Z_R^{[d]} \rightarrow B_R$ to form its restriction $Z_{R,\ell}^{[d]}$. Its abstract isomorphism class is independent of the choice of ℓ , and*

$$Z_{R,\ell}^{[d]} \simeq Z_{R',\ell'}^{[d]} \iff R' = gR \text{ for some } g \in \mathrm{PGL}(V).$$

The smallest uniform reconstruction order for these curve-supported neighbourhoods is again d . Their reductions all have dimension one, instead of $n(p-n)$.

Proof. The tautological bundle on a Schubert line is $\mathcal{O}_{\mathbb{P}^1}(-1) \oplus \mathcal{O}_{\mathbb{P}^1}^{n-1}$. Pulling back the homogeneous ideal of the graph neighbourhood gives exactly (8.2) with that bundle. Its isomorphism type depends on this bundle and R , not on the chosen inclusion of the line in the socle. The determinant degree is -1 , so Theorem 8.1 applies. Its lower-order schemes are all independent of the pencil, while $\langle x_1^2, x_1 x_2 \rangle$ and $\langle x_1^2, x_2^2 \rangle$ are inequivalent. This proves the uniform minimality assertion. $\square$

This restriction is made in the construction; it is not a claim that an abstract neighbourhood canonically chooses a Schubert line. No line is supplied to an isomorphism in the corollary: the reduced curve is recovered as the reduction of its source. Nor is the corollary a statement about one unmarked Artinian normal slice. The projective curve and the nontrivial ruling remain essential to this argument. The same theorem applies, for example, to $\mathcal{U} = \mathcal{O}_C(-P) \oplus \mathcal{O}_C^{n-1}$ on every smooth projective connected curve with a point P . It isolates the properness and ruling obstruction used in reconstruction, independently of the original Grassmannian realization.

9. RECONSTRUCTION OF MOVING PENCILS AND THEIR ISOMORPHISMS

The inverse need not be applied separately to constant pencils. A single unmarked finite scheme can retain a pencil varying over its entire projective reduction. We prove this assertion,

including the recovery of the normal bundle and the ambiguity in lifting the recovered data to an isomorphism. The coefficient lines in this section are subbundles; constancy of the pencil is not assumed.

9.1. The family and its first relation bundle. Fix $n \geq 3$, and write

$$N = \binom{n+1}{2}, \quad p = N - 2, \quad d = n + 2p = n^2 + 2n - 4, \quad M = \binom{N}{2}.$$

An admissible datum consists of a smooth connected projective complex variety B , a rank- n vector bundle $\mathcal{U}$ on B , a constant n -dimensional vector space V , and a rank-two locally direct-summand subbundle

$$\mathcal{R} \subset \mathrm{Sym}^2 V \otimes \mathcal{O}_B.$$

We require that $\mathbb{P}_B(\mathcal{U}^*)$ is not a trivial projective bundle. Put $\mathcal{S} = (\mathrm{Sym}^2 V \otimes \mathcal{O}_B)/\mathcal{R}$ and $\mathcal{E} = \mathrm{Hom}(\mathcal{U}, V \otimes \mathcal{O}_B)$. On its total space, let T be the universal map and let γ denote the quotient onto $\mathcal{S}$. Define the homogeneous ideal and its truncations by

$$\begin{aligned} \mathcal{I}_{\mathcal{R}} &= (\det T) I_p(\gamma \mathrm{Sym}^2 T), \\ \mathcal{A}_{\mathcal{R}}^{[k]} &= \mathrm{Sym}(\mathcal{E}^*)/(\mathcal{I}_{\mathcal{R}} + \mathfrak{a}^{k+1}), \quad Y_{\mathcal{R}}^{[k]} = \mathrm{Spec}_B \mathcal{A}_{\mathcal{R}}^{[k]}, \end{aligned} \tag{9.1}$$

where $\mathfrak{a}$ is the positive-degree ideal. Determinant-line changes of frame multiply local generators by units, so this is an ideal sheaf, independent of all frames. This construction is the pullback of the coefficient construction in Theorem 12.1, with an arbitrary rank- n source bundle. It is not necessary to choose an embedding of B in a socle Grassmannian.

Lemma 9.1 (A varying coefficient line). *The ideal $\mathcal{I}_{\mathcal{R}}$ has no component of degree less than d and is generated by its degree- d component $\mathcal{K}_{\mathcal{R}}$. This component is a subbundle of $\mathrm{Sym}^d \mathcal{E}^*$ of rank M . After cancellation of the determinant line, its degree- $2p$ coefficient module determines $\mathcal{R}$ as a subbundle, once the two tensor factors of $\mathcal{E}$ are supplied. In particular, $Y_{\mathcal{R}}^{[d]} \rightarrow B$ is finite locally free of rank*

$$\binom{n^2 + d}{d} - M.$$

Proof. Put $F(H) = \bigwedge^p \mathrm{Sym}^2 H$ for rank- n vector spaces or bundles H . This is an irreducible polynomial representation by Lemma 6.2; its dimension is M . Write $\mathcal{L}_{\mathcal{R}} = \bigwedge^p \mathcal{S}^*$, viewed, by $\bigwedge^p \gamma^*$, as a line subbundle of $F(V)^* \otimes \mathcal{O}_B$. The residual minors are precisely the image of

$$\mathcal{L}_{\mathcal{R}} \otimes F(\mathcal{U}) \longrightarrow \mathrm{Sym}^{2p}(V^* \otimes \mathcal{U}), \quad \beta \otimes \xi \longmapsto [T \mapsto \beta(F(T)\xi)]. \tag{9.2}$$

The full map $F(V)^* \otimes F(U) \rightarrow \mathrm{Sym}^{2p}(V^* \otimes U)$ is the nonzero inclusion of its unique Cauchy summand: evaluation at $T = 1$ is nonzero, as in (6.6). Restricting to a nonzero line on the left therefore has rank M in every fibre. The coefficients vary algebraically. The constant-rank criterion makes the image a subbundle; equivalently one may choose a nonvanishing maximal minor of this coefficient map locally on the Grassmannian of pencils and pull it back to B . Multiplication by the determinant, a nonzero polynomial in every fibre, again has constant rank M .

All defining generators have degree d , proving the assertion about the ideal. For the recovery assertion, the residual image, inside the indicated Cauchy summand, is $\mathcal{L}_{\mathcal{R}} \otimes F(\mathcal{U})$. Contraction with $F(\mathcal{U})^*$ has image exactly $\mathcal{L}_{\mathcal{R}}$ as a line subbundle of $F(V)^* \otimes \mathcal{O}_B$. This statement is an equality of images of bundle maps, not just equality at closed points. Exterior duality identifies it with the Pluecker line $\bigwedge^2 \mathcal{R}$, up to the fixed determinant line of $\mathrm{Sym}^2 V$. The Grassmannian Pluecker embedding consequently recovers the morphism $B \rightarrow \mathrm{Gr}(2, \mathrm{Sym}^2 V)$, and its pullback of the tautological bundle recovers $\mathcal{R}$. Finally, the truncated algebra is the sum of its free graded

pieces below d and the locally free degree- d quotient by $\mathcal{K}_{\mathcal{R}}$. Its rank is the asserted binomial sum minus M . $\square$

9.2. The unmarked inverse.

Theorem 9.2 (Reconstruction of a moving pencil). *Let $(B_i, \mathcal{U}_i, V_i, \mathcal{R}_i)$, $i = 1, 2$, be admissible data of the same rank n . Then*

$$Y_{\mathcal{R}_1}^{[d]} \simeq Y_{\mathcal{R}_2}^{[d]} \quad (9.3)$$

as abstract complex schemes if and only if there exist an isomorphism $\sigma : B_1 \rightarrow B_2$, a constant linear isomorphism $g : V_1 \rightarrow V_2$, and a bundle isomorphism $h : \mathcal{U}_1 \rightarrow \sigma^ \mathcal{U}_2$ such that*

$$\sigma^* \mathcal{R}_2 = (\text{Sym}^2 g) \mathcal{R}_1. \quad (9.4)$$

The reduced base, normal tensor orientation, isomorphism class of the actual source bundle, and the varying pencil are recovered together. Neither the projection onto the base nor the ambient bundle is supplied in (9.3). The constant left map cannot be replaced by a separately chosen change of coordinates in each fibre. A pair (g, h) defining the same normal-bundle map is unique up to simultaneous multiplication by a nonzero constant.

Proof. Let $\mathfrak{n}_i$ be the nilradical of $\mathcal{O}_{Y_{\mathcal{R}_i}^{[d]}}$. Since the positive-degree ideal is nilpotent and B_i is reduced, the reduction is exactly B_i , and this ideal is exactly $\mathfrak{n}_i$. There are no relations below degree d , so multiplication intrinsically recovers

$$\mathcal{E}_i^* = \mathfrak{n}_i / \mathfrak{n}_i^2, \quad \mathcal{K}_i = \ker \left[\text{Sym}^d(\mathfrak{n}_i / \mathfrak{n}_i^2) \longrightarrow \mathfrak{n}_i^d \right]. \quad (9.5)$$

The map is well defined since $\mathfrak{n}_i^{d+1} = 0$. An abstract isomorphism gives σ , an isomorphism of these normal bundles, and the matching relation subbundles. In particular it gives an isomorphism of the untruncated homogeneous cones

$$\text{Spec}_{B_i}(\text{Sym } \mathcal{E}_i^* / (\mathcal{K}_i)).$$

This cone is reconstructed from the first relation module; it is not the associated graded algebra of the finite scheme, whose degrees above d vanish.

The reduction of this cone is the determinant cone. Indeed the residual quadratic map is surjective wherever T is invertible, whereas the determinant factor annihilates the ideal wherever T is singular. On a frame chart the determinant hypersurface is reduced over the smooth base, so the radical ideal is precisely the determinant ideal. Successive relative reduced singular loci give the lower matrix-rank cones. The rank-one cone has the two ruling parameter schemes

$$\mathbb{P}(V_i) \times B_i, \quad \mathbb{P}_{B_i}(\mathcal{U}_i^*),$$

by Lemma 3.1. Only the first is projectively trivial. Thus an isomorphism cannot interchange them. The induced map on the trivial ruling is a morphism $B_1 \rightarrow \text{Isom}(\mathbb{P}(V_1), \mathbb{P}(V_2))$. After choosing one identification this target is the affine variety PGL_n . Since B_1 is connected and projective, every such morphism is constant. Choose one constant linear lift g .

We check that no line-bundle ambiguity remains in the other factor. Locally trivialize both source bundles. The linear rank-one preserver with its left projective map fixed to $[g]$ is of the form

$$T \longmapsto gTh^{-1} \quad (9.6)$$

by Lemma 3.2. Once this particular g is fixed, the local linear map h is unique: if two right maps give the same value for every T , they are equal. The local maps therefore agree on overlaps and glue to an actual bundle isomorphism $\mathcal{U}_1 \rightarrow \sigma^* \mathcal{U}_2$. Alternatively, after removing g the normal-bundle map commutes with $\text{End}(V_1)$; the commutant is the right-factor homomorphism

bundle. Rescaling the constant lift g rescales h by the same constant. This proves both existence and the asserted uniqueness for a given normal-bundle map.

Cancel the intrinsic determinant ideal in the recovered homogeneous ideal. On a frame chart the determinant polynomial is a non-zero-divisor and $((\det T)\mathcal{J} : (\det T)) = \mathcal{J}$; the identities glue without selecting a global determinant generator. Apply Lemma 9.1 to the degree- $2p$ part. The right map h acts on the full factor $F(\mathcal{U})$, so contraction against its dual removes that change of frame. The remaining line subbundle transforms by the one constant left map g . Exterior duality yields exactly (9.4). This uses the subbundle version proved above, not the constant-coefficient hypothesis in Theorem 5.1.

Conversely σ, g, h satisfying (9.4) identify the quotient bundles $\mathcal{S}_i$, the quadratic maps, and their maximal-minor ideals. The determinant changes by a unit in local frames. The map (9.6) consequently identifies both the homogeneous ideals and every truncation, proving (9.3). $\square$

Corollary 9.3 (Uniform sharpness over an admissible base). *For fixed $B, \mathcal{U}, V$, all schemes $Y_{\mathcal{R}}^{[k]}$ with $k < d$ are independent of $\mathcal{R}$. Order d is the smallest uniform order at which the moving pencil is recoverable in the sense of Theorem 9.2.*

Proof. Below d the ideal $\mathcal{I}_{\mathcal{R}}$ contributes no relations, and the algebra is $\bigoplus_{j=0}^k \text{Sym}^j \mathcal{E}^*$. Constant pencils are among the admissible subbundles. The pencils $\langle x_1^2, x_1 x_2 \rangle$ and $\langle x_1^2, x_2^2 \rangle$ are inequivalent: all members of the first have a common linear factor and those of the second do not. A base automorphism does not change either constant pencil. The theorem separates their order- d schemes, whereas every lower truncation agrees. This proves uniform, not pairwise, minimality. $\square$

9.3. The part of an isomorphism that the inverse does not record. The recovered data classify isomorphism classes, but an isomorphism of finite schemes need not itself be linear in the nilpotent directions. The following statement identifies this distinction without replacing it by a claim about moduli stacks.

Fix an admissible datum, put $Y = Y_{\mathcal{R}}^{[d]}$, and let $\mathcal{L}_{\mathcal{R}}$ be the group of triples (σ, g, h) satisfying (9.4), modulo $(g, h) \sim (cg, ch)$ for $c \in \mathbb{C}^*$. Its group law is composition of their induced normal-bundle maps. Write $\text{Aut}_{\mathbb{C}}(Y)$ for the group of complex scheme automorphisms. For $j \geq 1$, define G_j to consist of the automorphisms inducing the identity on the reduction and satisfying

$$(\alpha^* - 1)(\mathfrak{n}^k) \subseteq \mathfrak{n}^{k+j} \quad \text{for every } k \geq 0, \quad \mathfrak{n}^0 = \mathcal{O}_Y. \quad (9.7)$$

They act on sheaves on the same underlying topological space.

Theorem 9.4 (Linear data and the nilpotent kernel). *There is an exact sequence of groups*

$$1 \longrightarrow G_1 \longrightarrow \text{Aut}_{\mathbb{C}}(Y) \longrightarrow \mathcal{L}_{\mathcal{R}} \longrightarrow 1. \quad (9.8)$$

The displayed homogeneous presentation of Y supplies a splitting. Moreover $G_{d+1} = 1$, $[G_i, G_j] \subseteq G_{i+j}$, and there are natural injections

$$G_j/G_{j+1} \hookrightarrow H^0(B, \text{Der}_{\mathbb{C}}(\text{gr}_{\mathfrak{n}} \mathcal{O}_Y)_j) \quad (9.9)$$

into additive groups of degree- j derivations. Thus the kernel is nilpotent and records precisely the automorphisms invisible on the associated graded scheme. No surjectivity onto the derivation spaces is asserted.

Proof. An automorphism induces a base map and a degree-one bundle map. The proof of Theorem 9.2 identifies these with a unique class of triples in $\mathcal{L}_{\mathcal{R}}$. The graded algebra is generated in degree one over degree zero, with relation subbundle $\mathcal{K}_{\mathcal{R}}$. Hence its entire graded automorphism is already determined by that triple. The kernel is the identity on every graded piece, which is exactly condition (9.7) for $j = 1$. Conversely the homogeneous maps $T \mapsto gTh^{-1}$ give actual automorphisms of Y , respect composition, and induce the prescribed triples. This proves exactness

and the splitting. A choice of the homogeneous presentation supplies that section; the assertion does not endow an arbitrary unmarked presentation with a preferred section.

For $\alpha \in G_j$, put $D_\alpha = \alpha^* - 1$. It raises filtration by j , and is therefore nilpotent as an additive operator. The inverse of $1 + D_\alpha$ is the finite geometric series. Composition of filtration-raising operators shows that a commutator of elements of G_i, G_j raises filtration by at least $i + j$. Also $G_{d+1} = 1$ follows by taking $k = 0$, since $\mathfrak{n}^{d+1} = 0$.

The leading term of D_α defines maps $\mathfrak{n}^k / \mathfrak{n}^{k+1} \rightarrow \mathfrak{n}^{k+j} / \mathfrak{n}^{k+j+1}$. They are independent of lifts. The identity

$$D_\alpha(ab) = aD_\alpha(b) + D_\alpha(a)b + D_\alpha(a)D_\alpha(b)$$

reduces to the Leibniz identity in the indicated graded degree, since $2j \geq j + 1$. Thus the leading term is a sheaf derivation over $\mathbb{C}$, including its action on degree zero; it need not be an $\mathcal{O}_B$ -linear derivation. Leading terms add under composition, and their kernel is exactly G_{j+1} . They glue globally, proving (9.9) and all assertions. $\square$

Example 9.5 (A nontrivial invisible automorphism). Let $B = \mathbb{P}^1$ and $\mathcal{U} = \mathcal{O}(-1) \oplus \mathcal{O}^{n-1}$. The bundle $\mathcal{E}^* = V^* \otimes \mathcal{U}$ has at least two trivial direct summands. Choose their degree-one generators x, y . The substitution

$$x \mapsto x + y^2$$

fixing y , the other summands and $\mathcal{O}_B$, defines a nonidentity automorphism of $Y_{\mathcal{R}}^{[d]}$ in G_1 , for every admissible $\mathcal{R}$. Indeed each relation has degree d ; its change under this substitution has degree at least $d + 1$ and vanishes in the truncation. The inverse is $x \mapsto x - y^2$. The automorphism is nonidentity because there are no degree-two relations. Therefore identifying all finite-scheme automorphisms with the stabilizer of the pencil would be false, even in this simple case. The inverse theorem and the exact sequence retain, rather than suppress, this higher-order freedom.

9.4. A nonconstant family across spectral collisions.

Example 9.6. Take $n = 4$, $B = \mathbb{P}_{[s:t]}^1$, $\mathcal{U} = \mathcal{O}(-1) \oplus \mathcal{O}^3$, and $V = \mathbb{C}^4$. In diagonal quadratic coordinates set

$$A = \text{diag}(1, 1, 1, 1), \quad B_{s,t} = \text{diag}(0, s, t, 2s + 3t).$$

The map $\mathcal{O} \oplus \mathcal{O}(-1) \rightarrow \text{Sym}^2 V \otimes \mathcal{O}$ with columns $A, B_{s,t}$ has rank two everywhere. The first diagonal entry shows that any scalar relation involving A has zero coefficient on A ; the next two entries then force $s = t = 0$, which is impossible on $\mathbb{P}^1$. Its image $\mathcal{R}$ is therefore a locally direct-summand pencil bundle. Since it contains the nondegenerate form A , it is regular at every point, including all spectral collisions.

On the open set of four distinct eigenvalues, an ordered cross-ratio of $0, s, t, 2s + 3t$ is

$$\chi = \frac{t(s + 3t)}{(2s + 3t)(t - s)}.$$

The rational function $J = (\chi^2 - \chi + 1)^3 / (\chi^2(1 - \chi)^2)$ is invariant under the six cross-ratio transformations arising from permutations of four points. It is nonconstant: at $[s : t] = [1 : 2]$ and $[1 : 3]$, the values of χ are $7/4$ and $15/11$, and the corresponding values of J differ. Thus the pencil family is not projectively isotrivial, even allowing a change of parameter within each pencil. The discriminant of $\det(zA - B_{s,t})$ is

$$4s^2t^2(2s + 3t)^2(s - t)^2(s + 3t)^2(s + t)^2. \quad (9.10)$$

No factor is inverted in forming $Y_{\mathcal{R}}^{[20]}$. This finite scheme is locally free over its entire projective reduction of rank $\binom{36}{20} - 45$. By Theorem 9.2, its unmarked isomorphism class recovers the whole morphism into the pencil Grassmannian, including the six collision points and the pencil there, up to one constant transformation of V and an automorphism of the base. This is a consequence

about a genuinely moving family, not just the pointwise Pluecker encoding. The cross-ratio and discriminant calculations are classical elementary readouts; their role here is to exhibit a family to which the unmarked global inverse applies.

Remark 9.7 (Scope of the family statement). The reduction in this section is smooth, projective and reduced. The source projective bundle is assumed nontrivial, which orients the two rulings. The theorem allows all fibre pencils, singular or regular; regularity is used only in the spectral description of the last example. Theorem 12.1 separately allows arbitrary complex base change with its specified relative socle ideal. The present intrinsic inverse does not replace that ideal by an absolute nilradical over a nonreduced base. Nor does (9.8) assert an equivalence of unmarked moduli stacks or a description of every infinitesimal deformation. It is an exact classification of geometric isomorphisms, their graded data, and the residual automorphism kernel for the stated admissible objects.

10. COEFFICIENT-SUPPORT ORIENTATION AND A LOCAL ALGEBRAIC INVERSE

For pencils there is a second, entirely local way to distinguish the two tensor rulings. It uses the asymmetry of the residual coefficient module rather than the topology of a projective bundle. Consequently the nontrivial-projectivization hypothesis in the preceding family theorem can be removed, and a single unmarked Artin local algebra already recovers the pencil.

10.1. The coefficient module orients its own tensor factors. If $K \subset A \otimes B$ is a linear subspace, define its two factor supports by

$$s_A(K) = \text{im}(K \otimes B^* \longrightarrow A), \quad s_B(K) = \text{im}(K \otimes A^* \longrightarrow B).$$

The maps here are contractions. Their ranks are invariant under separate changes of basis and are exchanged by interchanging the factors. In particular, if $K = L \otimes B$, then $s_A(K) = L$ and $s_B(K) = B$. The support ranks are therefore $(\dim L, \dim B)$, not the common dimension $\dim K$.

Lemma 10.1 (Asymmetric coefficient support). *Suppose a homogeneous ideal in the matrix space $\text{Hom}(U, V)$, with $\dim U = \dim V = n \geq 3$, has reduced support the determinant cone. Suppose its intrinsic residual ideal after determinant cancellation has, in one multiplicity-one Cauchy summand of degree q , the coefficient subspace*

$$L \otimes \mathbb{S}_\lambda U \subset \mathbb{S}_\lambda V^* \otimes \mathbb{S}_\lambda U, \quad 0 < \dim L < \dim \mathbb{S}_\lambda U.$$

A linear isomorphism carrying this ideal to another ideal with a coefficient subspace of the same displayed type cannot interchange the two matrix rulings. The conclusion is unchanged by line twists of the locally chosen tensor factors. It also applies fibrewise to subbundles of this type over a reduced base.

Proof. A linear isomorphism of determinant cones preserves their successive rank loci and therefore the rank-one cone. Its action on the two ruling components either preserves them or interchanges them. To make the familiar two alternatives explicit, identify the rank-one projectivization with $\mathbb{P}(V) \times \mathbb{P}(U^*)$. An automorphism preserving its two families of maximal linear spaces acts separately on the two projective factors. Each factor map is projective linear, by restricting the ambient linear map to a ruling. Removing their linear lifts leaves a map fixing every decomposable line; sums of tensors with one factor in common show that this remaining map is scalar. The preserving alternative is thus a left-right matrix map. In the interchanging alternative, compose with transposition to obtain the preserving alternative.

The determinant ideal is carried to the determinant ideal, so its ideal quotient is carried to the residual ideal. The Cauchy decomposition is functorial for separate changes of matrix factors; transposition exchanges the factors in its λ -summand. Put $m = \dim \mathbb{S}_\lambda U$, and $r = \dim L < m$.

The displayed subspace has support ranks (r, m) . Separate factor maps preserve these ranks, while transposition changes them to (m, r) . An ideal of the prescribed type on the other side has support ranks (r', m) with $r' < m$. It cannot be the transposed image. This excludes interchanging the rulings.

Tensoring a factor by a line changes its Schur module by a line power, not its contraction rank. The argument therefore does not require a specified linear lift of either projective factor. For bundle maps it applies to each geometric fibre; an interchange on any fibre would contradict the same strict rank inequality there. $\square$

For pencils, take $F(H) = \bigwedge^{N-2} \text{Sym}^2 H$. Lemma 6.2 makes this a single irreducible Schur module, of dimension $M = \binom{N}{2} > 1$. The residual module is precisely $\mathcal{L}_{\mathcal{R}} \otimes F(\mathcal{U})$, by Lemma 9.1. Its support ranks are $(1, M)$. The relevant strict inequality holds for every pencil, including singular pencils; it is not a generic-rank condition.

Theorem 10.2 (Removal of the projective-bundle restriction). *In Theorem 9.2, the hypothesis that $\mathbb{P}_B(\mathcal{U}^*)$ is nontrivial can be omitted. The coefficient-subbundle reconstruction, the smallest uniform order, and the automorphism exact sequence remain valid for every rank- n source bundle on a smooth connected projective complex base. In particular the base is allowed to be a point.*

Proof. The first-relation construction, the recovered homogeneous cone, and cancellation of the determinant use neither projective nontriviality nor positive dimension of the base. They recover both the unordered rulings and the residual coefficient module before a ruling is chosen. At this stage apply Lemma 10.1 with support ranks $(1, M)$. It excludes the transposed alternative and hence orients the left ruling even when both ruling bundles are trivial. This replaces exactly the nontrivial-projectivization step in the proof of Theorem 9.2.

The left ruling is the trivial bundle $\mathbb{P}(V) \times B$. Projectivity and connectedness still make its projective-matrix map constant. Choose its constant linear lift; uniqueness of the right bundle map on overlaps gives the same actual isomorphism h , and coefficient contraction recovers the entire pencil subbundle. The converse uses the same homogeneous map. The lower-order algebras and the two inequivalent constant pencils in Corollary 9.3 are unchanged. Finally, the proof of Theorem 9.4 uses only the resulting classification of graded data, not the earlier way of orienting the rulings, so it too applies without the restriction. $\square$

10.2. Every pencil from one Artin local algebra. Choose n -dimensional vector spaces U, V . Let $T = (t_{ij})$ be the universal matrix, and put $\mathfrak{m} = (t_{ij})$. For a pencil $R \subset \text{Sym}^2 V$, let γ_R be its quotient map and set

$$\mathfrak{A}_R^{[k]} = \mathbb{C}[t_{ij}] / ((\det T) I_p(\gamma_R \text{Sym}^2 T) + \mathfrak{m}^{k+1}). \quad (10.1)$$

An isomorphism below means an isomorphism of complex algebras, without a grading, tangent coordinates, matrix decomposition or chosen generators.

Theorem 10.3 (Unmarked local reconstruction). *For every $n \geq 3$ and every pair of quadratic pencils,*

$$\mathfrak{A}_R^{[d]} \simeq \mathfrak{A}_{R'}^{[d]} \iff R' = (\text{Sym}^2 g)R \text{ for some } g \in \text{GL}(V). \quad (10.2)$$

The algebra has embedding dimension n^2 and length $\binom{n^2+d}{d} - \binom{N}{2}$. Every lower-order algebra $\mathfrak{A}_R^{[k]}$, $k < d$, is independent of R . Thus $d = n^2 + 2n - 4$ is the smallest uniform reconstruction order already among these single-point objects.

Proof. Theorem 10.2 applies with $B = \text{Spec } \mathbb{C}$ and the bundle U . To spell out what its intrinsic construction uses in this case, the unique maximal ideal $\mathfrak{n}$ of the finite algebra determines its

tangent space and the multiplication kernel

$$K_R = \ker \left[\text{Sym}^d(\mathfrak{n}/\mathfrak{n}^2) \longrightarrow \mathfrak{n}^d \right].$$

This is precisely the degree- d ideal in the chosen presentation; nonlinear changes of generators contribute only terms above degree d to a degree- d relation. Hence an ungraded algebra isomorphism identifies the homogeneous cones generated by these kernels. Their determinant reductions recover the unordered matrix factors. The residual support ranks $(1, M)$ rule out transposition, and the remaining left-right map carries the left coefficient line to the left coefficient line. Exterior duality then gives $R' = (\text{Sym}^2 g)R$. Conversely such a map, together with any chosen right identification, induces the algebra isomorphism. There are no relations below degree d , so the embedding dimension and the lower-order assertion are immediate. The relation subspace at degree d has dimension M , giving the asserted length. The inequivalent common-factor and no-common-factor pencils prove uniform minimality as before. $\square$

In particular the earlier Schubert-line construction is a sufficient geometric realization, not a lower bound on the dimension of a supporting reduced scheme. The local inverse just proved retains no curve at all. Its additional ingredient is the coefficient-support asymmetry, which was not used in the earlier geometric orientation argument. This is a stronger inverse statement, not a conclusion from knowing the discriminant or the reduced determinant cone.

10.3. The local automorphism kernel is explicit. Put $a = n^2$, $\ell = \text{length } \mathfrak{A}_R^{[d]}$, and $G_R = \{g \in \text{GL}(V) : (\text{Sym}^2 g)R = R\}$. The symbol $\mathfrak{n}^2$ in the next statement is the square of the maximal ideal of the finite algebra.

Proposition 10.4 (Tangent-identity automorphisms). *There is a split exact sequence of complex automorphism groups*

$$1 \longrightarrow U_R \longrightarrow \text{Aut}_{\mathbb{C}}(\mathfrak{A}_R^{[d]}) \longrightarrow (G_R \times \text{GL}(U))/\mathbb{C}^* \longrightarrow 1, \quad (10.3)$$

where the scalar subgroup is diagonal. The kernel U_R is a unipotent algebraic group whose underlying variety is an affine space of dimension

$$a(\ell - 1 - a).$$

More explicitly, after choosing the a degree-one generators $x_1, \dots, x_a$, its elements are exactly the substitutions $x_i \mapsto x_i + u_i$ with arbitrary $u_i \in \mathfrak{n}^2$. The affine-space parametrization is not an assertion that this group law is additive.

Proof. The local inverse and the support-rank argument identify the linear stabilizer of K_R with the indicated left-right quotient; transposition is absent. These linear maps lift homogeneously, giving the quotient and splitting. An automorphism in the kernel is the identity on $\mathfrak{n}/\mathfrak{n}^2$, and consequently has exactly the displayed generator images.

Conversely choose arbitrary $u_i \in \mathfrak{n}^2$. Under this substitution a homogeneous defining relation of degree d changes only by terms in $\mathfrak{n}^{d+1} = 0$, so all defining relations are respected. The induced map is the identity on the associated graded algebra, hence is invertible as a finite filtered linear map and therefore as an algebra map. There are no further conditions on the u_i . Reading the generator images and performing substitutions are polynomial mutually inverse maps between the kernel and the vector space $(\mathfrak{n}^2)^{\oplus a}$. Since $\dim \mathfrak{n}^2 = \ell - 1 - a$, this proves the variety and dimension assertions. On a basis ordered by degree every such automorphism is upper unitriangular; the algebraic group is therefore unipotent. Its law includes higher substitution terms, which accounts for its generally noncommutative structure. $\square$

The positive-dimensional family statement remains useful after the local inverse: it recovers an entire varying subbundle and the actual source bundle up to one global left transformation, whereas separate fibrewise local isomorphisms need not glue. The geometric automorphism statements here still do not identify unmarked moduli stacks or all deformations over nonreduced parameter bases.

11. SPECTRAL TORSION AND RANK-PRESERVING SPECIALIZATION

The finite-neighbourhood theorem recovers more than the discriminant or the corank of each member of a regular pencil. We identify its classical spectral invariant and then give a flat specialization on which the entire discriminant and every reduced rank locus remain constant, while the recovered invariant changes. The two specializations are separated at exactly the order of Theorem 7.1. The parameter space throughout is the usual Grassmannian of pencils; the failure map is the original multiplication map, not an added coefficient-evaluation block.

11.1. The spectral torsion sheaf. Let $R \subset \text{Sym}^2 V$ be a regular pencil, meaning that it contains a nonsingular quadratic form. On its parameter line $L_R = \mathbb{P}(R)$, the tautological quadratic form gives an exact sequence

$$0 \longrightarrow V^* \otimes \mathcal{O}_{L_R}(-1) \xrightarrow{q_R} V \otimes \mathcal{O}_{L_R} \longrightarrow \mathcal{T}_R \longrightarrow 0. \quad (11.1)$$

The first arrow is injective because its determinant is nonzero at the generic point of the smooth integral curve L_R . Thus $\mathcal{T}_R$ is a torsion sheaf of length n , whose zeroth Fitting divisor is the full discriminant. The pair $(L_R, \mathcal{T}_R)$ is considered up to isomorphism of the parameter line. Its definition involves no choice of a basis of R . The relation with elementary divisors is classical; see [4, §2, Corollary 2.1 and equation (5)]. We record the argument to make precise which datum is recovered.

Lemma 11.1 (Sheaves on the affine spectral line). *Let A, B be symmetric complex matrices with A invertible. On the affine line with coordinate z , the cokernel of $zA - B$ is the finite-dimensional $\mathbb{C}[z]$ -module on $\mathbb{C}^n$ for which multiplication by z is $C = A^{-1}B$. An isomorphism of such sheaves, in this fixed coordinate, is equivalent to similarity of the corresponding matrices.*

Proof. Multiplication on the target by A^{-1} identifies the presentation with $zI - C$. Reduction of a polynomial vector modulo these relations sends $\sum_j z^j v_j$ to $\sum_j C^j v_j$; constant vectors give the inverse identification. The equivalence between coherent sheaves on an affine scheme and finitely generated modules identifies a finite-length sheaf supported in this affine line with this module. The support is finite, so its underlying complex vector space is finite dimensional. A module isomorphism is a linear isomorphism G satisfying $GC = C'G$, which is exactly similarity. Extension by zero across a point outside the support loses no sheaf data. $\square$

Lemma 11.2 (Polynomial square roots of invertible operators). *For every invertible complex matrix K there is $h \in \mathbb{C}[x]$ with $h(K)^2 = K$. If K is self-adjoint for a nondegenerate symmetric form and commutes with C , then $h(K)$ has both properties.*

Proof. Write the minimal polynomial as $\prod_\lambda (x - \lambda)^{a_\lambda}$, where $\lambda \neq 0$. In its local Artinian factor choose a square root $\mu_\lambda^2 = \lambda$, and put

$$h_\lambda(x) = \mu_\lambda \sum_{j=0}^{a_\lambda-1} \binom{1/2}{j} \left(\frac{x - \lambda}{\lambda} \right)^j.$$

The formal binomial identity, truncated modulo $(x - \lambda)^{a_\lambda}$, gives $h_\lambda(x)^2 = x$. The Chinese remainder theorem joins these polynomials to a single class modulo the minimal polynomial.

Evaluation at K proves the assertion even when K is not semisimple. Polynomials in one self-adjoint operator remain self-adjoint, and every polynomial in K commutes with every operator commuting with K . $\square$

Theorem 11.3 (Spectral reconstruction at the first relation). *Let $n \geq 3$, $d = n^2 + 2n - 4$, and let R, R' be regular pencils. The following are equivalent:*

- (1) *the unmarked schemes $Z_R^{[d]}$ and $Z_{R'}^{[d]}$ are isomorphic;*
- (2) *there is an isomorphism $f : L_R \rightarrow L_{R'}$ with $\mathcal{T}_R \simeq f^* \mathcal{T}_{R'}$;*
- (3) *the pencils R, R' are projectively equivalent.*

In particular the finite scheme determines the full root-labelled spectral torsion sheaf, not only its Fitting divisor or its support. This is an equivalence of isomorphism classes of objects; it is not an assertion identifying their automorphism groups or moduli stacks.

Proof. The equivalence of the first and third conditions is Theorem 7.1. A congruence of pencils induces an isomorphism of (11.1), so the third implies the second.

For the converse choose a common affine coordinate whose point at infinity avoids the two spectral supports, using f to identify the lines. Write the corresponding symmetric pairs as (A, B) and (A', B') , with A, A' invertible, and set $C = A^{-1}B$, $C' = A'^{-1}B'$. Choose the presentation $zA - B$ for the affine coordinate on both lines. Lemma 11.1 identifies the two sheaves with the modules of C, C' , so their isomorphism gives a similarity. After applying that change of coordinates, assume $C' = C$. Both A and A' are nondegenerate symmetric forms for which C is self-adjoint. Then $K = A^{-1}A'$ commutes with C and is self-adjoint for A . Lemma 11.2 provides a polynomial h with $h(K)^2 = K$. Put $H = h(K)$. It commutes with C , is self-adjoint for A , and satisfies

$$H^t A H = A', \quad H^t B H = H^t A C H = A' C = B'.$$

Thus the pairs, and hence the pencils, are congruent. This also shows why the spectral sheaf alone suffices here; no additional pairing on that sheaf is required over $\mathbb{C}$. $\square$

For $x \in \text{Supp } \mathcal{T}_R$, put

$$\ell_a(x) = \text{length}(\mathcal{T}_{R,x}/\mathfrak{m}_x^a \mathcal{T}_{R,x}), \quad a \geq 0. \quad (11.2)$$

If the elementary-divisor exponents at x are $e_1, \dots, e_b$, the structure theorem over the discrete valuation ring $\mathcal{O}_{L_R,x}$ gives

$$\ell_a(x) = \sum_{j=1}^b \min(a, e_j), \quad \ell_a(x) - \ell_{a-1}(x) = \#\{j : e_j \geq a\}. \quad (11.3)$$

Thus these intrinsic local lengths recover every exponent, including repeated roots. The corank of the pencil at x gives only $\ell_1(x)$, and the multiplicity of the discriminant gives only the eventual value $\ell_a(x) = \sum_j e_j$. The intermediate values are genuinely additional information.

11.2. A specialization with constant discriminant and coranks. The following family keeps both ends of this length profile fixed. Fix $i \in \mathbb{C}$ with $i^2 = -1$, and set

$$N_0 = \begin{pmatrix} 1 & i \\ i & -1 \end{pmatrix}, \quad P_t = \frac{t}{2} N_0, \quad A_0 = \begin{pmatrix} 0 & I_2 \\ I_2 & 0 \end{pmatrix}, \quad C_t = \begin{pmatrix} P_t & I_2 \\ 0 & P_t \end{pmatrix}. \quad (11.4)$$

Put

$$B_t = A_0 C_t = \begin{pmatrix} 0 & P_t \\ P_t & I_2 \end{pmatrix}, \quad R_t = \langle A_0, B_t \rangle \subset \text{Sym}^2 \mathbb{C}^4. \quad (11.5)$$

Both matrices are symmetric and A_0 is invertible. They are independent for every t . For $n > 4$, append I_{n-4} to A_0 and append $\text{diag}(\mu_5, \dots, \mu_n)$ to B_t , where the μ_j are fixed distinct nonzero numbers. Denote the resulting matrices by $A^{(n)}, B_t^{(n)}$ and their pencil again by R_t .

Theorem 11.4 (Separation inside fixed reduced rank data). *For every $n \geq 4$, the family R_t , $t \in \mathbb{A}^1$, has the following properties.*

- (1) *Its full discriminant, with multiplicities, is independent of t :*

$$\det(sA^{(n)} + uB_t^{(n)}) = s^4 \prod_{j=5}^n (s + \mu_j u). \quad (11.6)$$

For every minor size, the reduced determinantal locus on the marked parameter line is also independent of t .

- (2) *At the root $s = 0$, the spectral module has exponents $(3, 1)$ for $t \neq 0$ and $(2, 2)$ for $t = 0$. Consequently $Z_{R_t}^{[d]} \not\cong Z_{R_0}^{[d]}$ for every $t \neq 0$, whereas all the punctured fibres are isomorphic.*
- (3) *The schemes $Z_{R_t}^{[d]}$ form a flat projective family over $\mathbb{A}^1$, with fixed reduction. After one identification of the socle quotient bundles, every truncation of order $k < d$ is a constant family. Thus the first separating order remains exactly $d = n^2 + 2n - 4$ even under the constraints in part (1).*

Proof. The identities $N_0^2 = 0$ and $\text{rank } N_0 = 1$ give

$$C_t^2 = \begin{pmatrix} 0 & tN_0 \\ 0 & 0 \end{pmatrix}, \quad C_t^3 = 0, \quad C_t = \begin{pmatrix} I_2 \\ P_t \end{pmatrix} \begin{pmatrix} P_t & I_2 \end{pmatrix}. \quad (11.7)$$

The right factor in the last expression is surjective and the left factor is injective. Hence $\text{rank } C_t = 2$ for every t , whereas $\text{rank } C_t^2 = 1$ precisely when $t \neq 0$. The nilpotent partitions are therefore $(3, 1)$ and $(2, 2)$. Also $\det A_0 = 1$ and $\det(sI_4 + uC_t) = s^4$, proving (11.6). At $s = 0$ the full pencil has corank two for all t ; at each remaining root it has corank one, and elsewhere corank zero. Equality of these functions on $\mathbb{P}^1(\mathbb{C})$ implies equality of the reduced closed determinantal subschemes for each minor size.

At the unique multiplicity-four root, formula (11.3) yields respectively

$$(\ell_1, \ell_2, \ell_3, \dots) = (2, 3, 4, \dots), \quad (2, 4, 4, \dots).$$

No automorphism of the parameter line can exchange this root with a simple one. Theorem 11.3 separates the special fibre from the punctured fibres and identifies the latter with one another. This is a statement about geometric fibres; no choice of a simultaneous trivializing congruence on $\mathbb{A}^1 \setminus 0$ is needed.

For flatness we use the relative gluing of Theorem 12.1; the following coordinates identify its pullback in the present family. First note that $t \mapsto R_t$ is a morphism to $\text{Gr}(2, \text{Sym}^2 V)$, not merely a collection of pencils. In symmetric matrix coordinates the two fixed linear functionals

$$a(Q) = Q_{13} + iQ_{14}, \quad b(Q) = Q_{33}$$

satisfy $a(A^{(n)}) = b(B_t^{(n)}) = 1$ and $a(B_t^{(n)}) = b(A^{(n)}) = 0$. Their common kernel is a fixed complement to every R_t . It trivializes $S_{R_t} = \text{Sym}^2 V / R_t$, and hence identifies the relative socle Grassmannian with $B \times \mathbb{A}^1$.

On a frame chart of its tautological bundle, the graph description (7.4) identifies the truncated algebra with

$$\bigoplus_{j=0}^{d-1} \text{Sym}^j E^* \oplus (\text{Sym}^d E^* / K_{R_t}). \quad (11.8)$$

By Theorem 7.3, K_{R_t} is a subbundle of constant rank $\binom{N}{2}$, with locally free quotient. This statement commutes with pullback to the present base and with right changes of frame, so the local descriptions glue. Thus (11.8) is locally free on $B \times \mathbb{A}^1$, of rank $\binom{n^2+d}{d} - \binom{N}{2}$. The graph projection is a finite flat morphism to that base. Since B is projective, its composition

with projection to $\mathbb{A}^1$ is flat and projective. Below degree d there is no relation, and the fixed complement identifies the entire lower truncated algebra, including multiplication, with a constant family. These arguments prove flatness of the finite neighbourhoods; they make no assertion about flatness of the unrestricted failure schemes. $\square$

The qualifier “reduced” in part (1) is necessary. At $s = 0$, in a local parameter $z = s/u$, the gcd of the three-row minors of the four-dimensional block is a unit times z for $t \neq 0$ and a unit times z^2 for $t = 0$. Both radicals are (z) . Thus the family retains exactly the higher Fitting information discarded by reduction; it does not contradict the classical classification by the full determinantal ideals.

11.3. An independently visible change of reciprocal geometry. For a regular pencil let its reciprocal curve be the projective closure of the inverses of its nonsingular members. These curves and their strata are studied independently of multiplication-failure schemes in [4, §3].

Corollary 11.5 (A conic specializes to a line). *For the four-dimensional family (11.5), the reciprocal curve is a smooth plane conic when $t \neq 0$ and a projective line when $t = 0$. The unmarked twentieth neighbourhood distinguishes these two cases, while all earlier neighbourhoods, the full discriminant, and all reduced rank loci agree.*

Proof. For $s \neq 0$, multiplication on the right by the fixed invertible matrix A_0^{-1} identifies the projective inverse with

$$(sI_4 + uC_t)^{-1} = s^{-1}I_4 - us^{-2}C_t + u^2s^{-3}C_t^2.$$

If $t \neq 0$, the matrices I_4, C_t, C_t^2 are linearly independent: their span has dimension equal to the degree three of the minimal polynomial. The closure is therefore the quadratic Veronese image $[s : u] \mapsto [s^2 : -su : u^2]$, a smooth conic in its spanning plane. At $t = 0$ the last term vanishes, and I_4, C_0 are independent, giving a line. The remaining assertions follow from Theorem 11.4, with $d = 20$. The degree drop does not assert that the reciprocal curves themselves form a flat family; the flat family proved above consists of the finite failure neighbourhoods. $\square$

12. THE UNIVERSAL FINITE NEIGHBOURHOOD AND SPECTRAL STRATA

The finite-order construction is a family of actual neighbourhoods in multiplication-failure schemes. This fact is needed when passing from individual pencils to degenerations and infinitesimal parameter spaces. We establish it before discussing the spectral incidence schemes.

Put $P = \text{Gr}(2, \text{Sym}^2 V)$, and write $\mathcal{R}$ and $\mathcal{S} = (\text{Sym}^2 V \otimes \mathcal{O}_P)/\mathcal{R}$ for its universal relation and quotient bundles. Let

$$\pi : \mathcal{B} = \text{Gr}_P(n, \mathcal{S}) \longrightarrow P, \quad \mathcal{E} = \text{Hom}(\mathcal{U}, V \otimes \mathcal{O}_{\mathcal{B}}),$$

where $\mathcal{U}$ is tautological of rank n . As before, $N = n(n+1)/2$, $p = N - 2$, $d = n + 2p$, and $M = \binom{N}{2}$.

Theorem 12.1 (Universal neighbourhood and arbitrary base change). *The universal multiplication-failure scheme has a canonical order- d neighbourhood $\mathcal{Z}^{[d]}$ of $\mathcal{B}$. There is a rank- M vector subbundle $\mathcal{K} \subset \text{Sym}^d \mathcal{E}^*$ such that*

$$\mathcal{Z}^{[d]} = \text{Spec}_{\mathcal{B}} \left(\text{Sym} \mathcal{E}^* / (\mathcal{K}, \text{Sym}^{d+1} \mathcal{E}^*) \right). \quad (12.1)$$

This is finite locally free over $\mathcal{B}$, of rank

$$\binom{n^2 + d}{d} - \binom{N}{2}, \quad (12.2)$$

and is projective and flat over P . Its construction, its ideal-adic filtration, and its identification with the actual multiplication-failure neighbourhood commute with every morphism of complex schemes $T \rightarrow P$, including nonreduced ones. For $k < d$, its order- k truncation is $\mathrm{Spec}_{\mathcal{B}} \bigoplus_{j=0}^k \mathrm{Sym}^j \mathcal{E}^*$, with truncated multiplication. After trivializing $\mathcal{S}$ on the parameter base, these lower-order families are constant.

Proof. The locus in $\mathrm{Gr}_P(n, V \oplus \mathcal{S})$ on which projection to $\mathcal{S}$ is injective is canonically $\mathrm{Tot} \mathcal{E}$. It is an open neighbourhood of $\mathcal{B}$, not a choice of formal coordinates. On a frame chart of $\mathcal{U}$, let $T : \mathcal{U} \rightarrow V$ be the universal graph map and let $\gamma : \mathrm{Sym}^2 V \rightarrow \mathcal{S}$ be the quotient. The same block elimination that proves (7.4) works over its coordinate ring: the ideal of the multiplication-failure scheme is

$$(\det T) I_p(\gamma \mathrm{Sym}^2 T).$$

Every generator is homogeneous of degree d .

Here is also a coordinate-free description of its first relation bundle. The coefficient map, followed by determinant multiplication, is a morphism

$$\det V^* \otimes \det \mathcal{U} \otimes \det \mathcal{S}^* \otimes \bigwedge^p \mathrm{Sym}^2 \mathcal{U} \longrightarrow \mathrm{Sym}^d \mathcal{E}^*. \quad (12.3)$$

The inclusion used in this expression is $\det \mathcal{S}^* \hookrightarrow (\bigwedge^p \mathrm{Sym}^2 V)^*$. The determinant factors evaluate $\det T$, and the remaining factors evaluate $\bigwedge^p \mathrm{Sym}^2 T$ followed by the quotient. Thus the image is exactly the span of the displayed maximal minors, not a separately inserted coefficient module.

On a frame chart this map is the fixed injective linear coefficient map ν of Theorem 7.3, restricted to a line subbundle tensored with the fixed space $\bigwedge^p \mathrm{Sym}^2 \mathbb{C}^n$. Its image is a subbundle of rank M : locally split the line in the source, and split the fixed complex linear inclusion ν . Both splittings remain splittings over any complex algebra. Denote its image by $\mathcal{K}$. A right change of frame of $\mathcal{U}$ changes the graph coordinates and this coefficient map contragrediently. The determinant character in (12.3) is exactly the one arising from $\det T$. Hence these local ideals and their degree- d subbundles agree on overlaps. In particular, they glue as ideals of the given graph neighbourhood.

Let $\mathfrak{b}$ be the ideal of the zero section inside the multiplication-failure scheme. In graph coordinates its preimage is the positive-degree ideal. Quotienting by $\mathfrak{b}^{d+1}$ therefore gives precisely (12.1). This identifies the gluing with the actual ideal-adic neighbourhood, not merely with algebras having the correct geometric fibres. Its underlying module is

$$\bigoplus_{j=0}^{d-1} \mathrm{Sym}^j \mathcal{E}^* \oplus (\mathrm{Sym}^d \mathcal{E}^* / \mathcal{K}).$$

All terms are locally free. Counting monomials in n^2 variables gives (12.2). The morphism $\mathcal{B} \rightarrow P$ is smooth and projective, so the asserted flatness and projectivity follow. Finally the graph description, the split coefficient inclusion, and the homogeneous quotient all commute with base change. Below degree d the ideal contributes no relation. This proves the final assertion as well. $\square$

Remark 12.2. Over a nonreduced base, the theorem concerns the specified relative socle subscheme $\mathcal{B}_T$ and its ideal. It does not identify that ideal with the absolute nilradical of $\mathcal{Z}_T^{[d]}$, which can also contain nilpotents from the base. On geometric fibres $\mathcal{B}_t$ is exactly the reduction. Nor is this theorem an equivalence of unmarked deformation groupoids: extra automorphisms of truncated algebras are not being identified with pencil congruences.

12.1. Spectral incidence schemes from the first relation space. On the regular-pencil locus, pass locally to ordered generators (A, B) with A invertible, and put $C = A^{-1}B$. For a marked point with affine coordinate z , set $X = C - zI$. The next elementary presentation works over an arbitrary coefficient ring; no diagonalization or division by eigenvalue differences is involved.

Lemma 12.3 (Truncated spectral presentation). *Let D be a commutative ring, $X \in \text{Mat}_n(D)$, and $a \geq 1$. There is a natural isomorphism*

$$\text{coker}(wI - X) \otimes_{D[w]} D[w]/(w^a) \simeq \text{coker}(X^a : D^n \rightarrow D^n). \quad (12.4)$$

Equivalently, the an -square block presentation with diagonal blocks X and subdiagonal blocks $-I$ is equivalent, by invertible row and column operations over D , to $I_{n(a-1)} \oplus X^a$.

Proof. Map the class of $\sum_{j=0}^{a-1} w^j v_j$ to the class of $\sum_{j=0}^{a-1} X^j v_j$ modulo $X^a D^n$. The relation $wv = Xv$ and the relation $w^a v = 0$ show that this map is well-defined. Sending a constant vector to its class is inverse. Eliminating successively the blocks with coefficient $-I$ gives the stated presentation equivalence, using only unit pivots. This also proves naturality under ring homomorphisms. $\square$

Theorem 12.4 (Scheme-theoretic spectral readout). *In the fixed tensor presentation of Theorem 7.3, the first relation subspace determines, on the regular-pencil locus, every closed spectral incidence scheme*

$$\mathfrak{S}_{a,h} = V(I_{n-h+1}((C - zI)^a)), \quad 1 \leq a, h \leq n. \quad (12.5)$$

This determination commutes with arbitrary complex base change. At a geometric point, membership is equivalent to $\text{length}(\mathcal{T}_z/\mathfrak{m}_z^a \mathcal{T}_z) \geq h$. Consequently the schemes of specified spectral length profiles, and not only their reduced supports, can be read from the order- d relation subspace.

Proof. The closed immersion Φ of Theorem 7.3 has an inverse on its scheme-theoretic image. Thus a family of relation subspaces in that image recovers its morphism to P , over any base, and hence its universal pencil. In the displayed chart its spectral sheaf has presentation $(z+w)I - C = wI - X$. Lemma 12.3 gives the presentation X^a for restriction to the order- a fat point centred at z . Its $(h-1)$ -st Fitting ideal is exactly the ideal in (12.5); Fitting ideals of finitely presented modules commute with base change. On a field the cokernel has dimension at least h exactly when these minors vanish.

These descriptions are independent of the chosen frames. A change of basis in V conjugates X ; a change of affine coordinate replaces the parameter ideal at the marked point by a unit multiple locally. Alternatively, restriction of the spectral sheaf to the fat diagonal on its parameter line defines the same finite module without a coordinate. Thus the Fitting ideals glue. Rank equality is imposed by the corresponding rank inequality together with the open complement of the next inequality. Intersecting these locally closed conditions gives each specified length profile. Formula (11.3) then identifies its geometric fibres with the prescribed elementary-divisor partition. $\square$

12.2. Realizing the complete Segre stratification. Let P_σ denote the reduced closure in P of a regular Segre stratum, and let $J_\sigma = \Phi(P_\sigma)$. Because Φ is a closed immersion, it gives isomorphisms

$$P_\sigma \simeq J_\sigma, \quad P_{\sigma_1} \cap_P \cdots \cap_P P_{\sigma_s} \simeq J_{\sigma_1} \cap_{\Phi(P)} \cdots \cap_{\Phi(P)} J_{\sigma_s}, \quad (12.6)$$

where the intersections carry their scheme structures. In particular their local rings and formal completions are preserved. Pulling back Theorem 12.1 realizes these parameter schemes by flat projective families of the actual finite failure neighbourhoods. The classical inclusions among Segre-stratum closures [4, Theorem 5.1] are therefore all realized, rather than just the one degeneration of Theorem 11.4.

There are two distinct assertions here. The Segre closure order is classical, and transport of closed subschemes through a closed immersion is formal. The geometric input proved here is the base-change-compatible realization by actual multiplication-failure neighbourhoods and the recovery of their spectral Fitting schemes. The ambient parameter space in (12.6) is the image of Φ , not a moduli space of every possible unmarked deformation of a finite scheme.

13. FIXED-DISCRIMINANT STRATA AND ALL ADMISSIBLE SPECIALIZATIONS

The universal family of Section 12 and the example of Section 11 lead to a classification of the entire fixed-discriminant, fixed-reduced-rank locus. The spectral orbit order is classical. Every allowed specialization is realized here by a flat family of the original finite multiplication-failure neighbourhoods, and the same first relation detects every strict transition.

Fix distinct $\alpha_1, \dots, \alpha_s \in \mathbb{C}$, positive m_i with $\sum m_i = n \geq 3$, and $1 \leq c_i \leq m_i$. Fix a nonsingular symmetric A , and let $\mathfrak{p}_A$ be its self-adjoint endomorphisms. Consider the reduced locally closed variety

$$X_{\mathbf{m}, \mathbf{c}}(A) = \left\{ C \in \mathfrak{p}_A : \det(zI - C) = \prod_i (z - \alpha_i)^{m_i}, \quad \dim \ker(C - \alpha_i I) = c_i \right\}. \quad (13.1)$$

Exclude $s = 1, c_1 = n$, for which C is scalar and $\langle A, AC \rangle$ is not a pencil. Every other datum is realized. Put $R_C = \langle A, AC \rangle$, $d = n^2 + 2n - 4$, and $Z_C = Z_{R_C}^{[d]}$. The full binary determinant is $\det(A) \prod_i (u + \alpha_i v)^{m_i}$. The coranks and labelled roots specify all reduced determinantal loci on the line.

Write $\mathcal{P}(m, c)$ for partitions into exactly c positive parts. For equal-size partitions, $\mu \preceq \lambda$ means $\sum_{j \leq a} \mu_j \leq \sum_{j \leq a} \lambda_j$ for every a ; write λ' for the conjugate partition.

13.1. The classical orthogonal input. We isolate the orbit theorem used below, with its precise group and source. Let (H, A) be a nondegenerate complex symmetric space of dimension m . The acting group is the full orthogonal group $O(A)$, and the operators are A -self-adjoint, not elements of the skew-adjoint Lie algebra $\mathfrak{o}(A)$.

Lemma 13.1 (Self-adjoint nilpotent orbits). *The full $O(A)$ -orbits of self-adjoint nilpotent operators are indexed by all partitions λ of m . If $\mathcal{O}_\lambda$ denotes the orbit, then*

$$\dim \mathcal{O}_\lambda = \frac{1}{2} \left(m^2 - \sum_a (\lambda'_a)^2 \right), \quad \mathcal{O}_\mu \subset \overline{\mathcal{O}_\lambda} \iff \mu \preceq \lambda.$$

An $O(A)$ -orbit is one $SO(A)$ -orbit when λ has an odd part. When every part is even, it is the disjoint union of two $SO(A)$ -orbits, interchanged by an isometry of determinant -1 . The closure statement above is for the full orthogonal orbit and does not identify the closures of its two components.

Proof. The classification and closure assertions are [5, Proposition 1, p. 444; Theorem 1, p. 447], respectively, with $\varepsilon = 1$ and $G(V)$ as defined in [5, §1.1, p. 443]. Remark 1 on p. 444 specifies that all partitions occur; the order in §1.4 on p. 446 is precisely the row-dominance order used here. The proof of Theorem 1 occupies pp. 447–451. These references concern the full group, not its identity component.

For the dimension formula there is a short independent calculation. The trace pairing on $\text{End } H$ is nondegenerate on both its self-adjoint and skew-adjoint summands. For a self-adjoint nilpotent N , the commutator with N exchanges the summands, and the two resulting maps are adjoint up to sign for that pairing. They have equal rank. Consequently $\dim \text{GL}(H)N = 2 \dim O(A)N$, also the identity in [5, §2.4, Remark 8(i), p. 456]. The general linear centralizer has dimension

$\sum_{i,j} \min(\lambda_i, \lambda_j) = \sum_a (\lambda'_a)^2$, as follows by writing homomorphisms between the cyclic $\mathbb{C}[t]$ -modules of the Jordan decomposition. This proves the stated dimension.

We give the component argument explicitly. The good bases of [5, Lemma 1, p. 444] decompose the symmetric space orthogonally into cyclic Jordan blocks, each with a reversed-identity Gram matrix. If one block has odd size, its negative identity, extended by the identity on the other blocks, centralizes N and has determinant -1 . The centralizer then meets both components of $O(A)$, so the orbit is a single $SO(A)$ -orbit.

For the converse, write

$$E_k = \ker N^k / (\ker N^{k-1} + N \ker N^{k+1}).$$

The form induced by $A(u, N^{k-1}v)$ on E_k is nondegenerate and symmetric; this is immediate on the same cyclic blocks. A commuting isometry induces $g_k \in O(E_k)$. A filtration of the Jordan module by its cyclic lengths and powers of t gives

$$\det g = \prod_k (\det g_k)^k.$$

Indeed the multiplicity-space action of a length- k block occurs k times on the associated graded; the strictly triangular terms of a module automorphism do not change its determinant. If all lengths are even, this product is one. The centralizer lies in $SO(A)$, and the index-two quotient of $O(A)$ therefore gives exactly two $SO(A)$ -orbits. Since $SO(A)$ is connected, these are the connected components of the full orbit. This does not preclude their closures from meeting. $\square$

Theorem 13.2 (Classification and specialization at the first relation). *The following statements hold.*

- (1) *The $O(A)$ -orbits in $X_{m,c}(A)$ are indexed by $\lambda \in \prod_i \mathcal{P}(m_i, c_i)$, with dimensions*

$$\dim \mathcal{O}_\lambda = \frac{1}{2} \left(n^2 - \sum_i \sum_a (\lambda'_{i,a})^2 \right). \quad (13.2)$$

Unmarked isomorphism classes of Z_C are these tuples modulo the permutation group induced on the roots by projective transformations preserving their multiplicities and coranks.

- (2) *A specialization from λ to μ inside (13.1) exists if and only if*

$$\mu_i \preceq \lambda_i \quad \text{for every } i. \quad (13.3)$$

Every such pair is realized over a smooth pointed algebraic curve, after shrinking at the special point, with A fixed and with constant full determinant and reduced rank loci.

- (3) *Every such pencil family induces a flat projective family of Z_C . After shrinking the curve, every truncation of order $k < d$ is constant. If at least one dominance inequality is strict, the generic and special Z_C are nonisomorphic; their first distinguishing order is d .*
- (4) *The recovered spectral module at α_i has length profile*

$$L_{i,a} = \sum_j \min(a, \lambda_{i,j}). \quad (13.4)$$

Every $L_{i,a}$ can only increase in the stated specialization. All these inequalities together are equivalent to (13.3), giving an intrinsic numerical criterion for every admissible transition.

Proof. We use Lemma 13.1. Its full-orthogonal classification and dominance order have exactly the group and self-adjoint convention needed here. The comparison for Grassmannian Segre strata is [4, Theorem 5.1].

The primary spaces $V_i = \ker(C - \alpha_i I)^{m_i}$ are an orthogonal direct sum: the Chinese remainder idempotents are polynomials in the self-adjoint C , hence self-adjoint projections. Each restricted

form is nondegenerate, and $C|_{V_i} = \alpha_i I + N_i$, with N_i self-adjoint nilpotent. The prescribed corank is its number of Jordan blocks. The classical theorem gives the indexing. Conversely, a Jordan block is self-adjoint for the reversed-identity Gram matrix; an isometry with the chosen nondegenerate form realizes each partition.

The space of orthogonal decompositions has dimension $(n^2 - \sum_i m_i^2)/2$. Adding the nilpotent orbit dimensions proves (13.2). These are dimensions in the fixed- A matrix space, not in an unmarked moduli stack. Full orthogonal orbits need not be connected; no irreducibility assertion is needed. Theorem 11.3 identifies the unmarked neighbourhood with the pencil, and its spectral module determines the partition at each root. A projective change of the line can therefore do exactly the stated root permutations. For one or two roots the projective stabilizer may be infinite; only its finite permutation image is used.

For necessity, the same polynomial idempotents decompose a family with fixed characteristic polynomial into subbundles of ranks m_i . The rank of N_i^a cannot increase at specialization. Since $m_i - \text{rank } N_i^a = \sum_j \min(a, \lambda_{i,j})$, this gives the length inequalities. Summing the first a columns of conjugate Young diagrams shows that they are equivalent to dominance: conjugation reverses the dominance order.

For sufficiency, fix an orthogonal decomposition of (V, A) . The closure criterion puts a chosen representative of μ_i in the closure of the orbit for λ_i . Thus the special tuple lies in the closure of the product of the generic orbits. Choose an irreducible component of that closure containing the point and meeting the generic orbit product. Successive general hyperplane sections through the point, avoiding the closed complement, give an integral algebraic curve with generic point in that open orbit product. Normalize it and choose a point over the prescribed point. A normal complex curve is smooth. Remove the other points outside the generic orbit product. A constant curve suffices in the equality case. The resulting block operator $\bigoplus_i (\alpha_i I + N_i(t))$ is self-adjoint for the same A . Its determinant is unchanged, and its coranks agree generically and specially, hence throughout the shrunk curve. This gives the required specialization. It does not assert constancy of the full higher Fitting ideals.

For part (3), pull back Theorem 12.1 along the curve's morphism to $P = \text{Gr}(2, \text{Sym}^2 V)$. That theorem identifies the family with the actual ideal-adic neighbourhoods and proves flatness and projectivity under arbitrary base change. No inference from fibrewise isomorphisms is needed. An affine Grassmannian chart around the special pencil supplies a fixed complement to nearby R_C , identifying S_{R_C} , B , and E . There is no relation below degree d , so all lower neighbourhoods become constant. For a strict dominance transition, $\sum_{i,a} (\lambda'_{i,a})^2$ strictly increases; this sum is invariant under permutations of the decorated roots. The generic and special tuples therefore cannot be related by such a permutation. Part (1) separates their unmarked degree- d neighbourhoods. Flatness is asserted only for these finite neighbourhoods, not unrestricted failure schemes. $\square$

Corollary 13.3 (Exactly when the coarse data lose a labelled class). *The number of root-labelled congruence classes with the prescribed data is $\prod_i \#\mathcal{P}(m_i, c_i)$. It is one precisely when, for every i , either $c_i = 1$ or $m_i - c_i \leq 1$. The first possible ambiguity with fixed full determinant and all reduced rank loci occurs in dimension four, at $(m, c) = (4, 2)$, with partitions $(3, 1)$ and $(2, 2)$.*

Proof. Subtracting one from every part identifies $\mathcal{P}(m, c)$ with partitions of $m - c$ into at most c parts. There is one if $c = 1$ or $m - c \leq 1$. Otherwise $(m - c)$ and $(m - c - 1, 1)$ are distinct admissible partitions. The theorem gives the count and the minimal example. $\square$

13.2. The first relation as a standard graded Betti space. The finite invariant has a usual Hilbert-scheme interpretation. The first Betti space means the generator space in the first free module of a minimal graded resolution of the quotient, not the space of relations among generators of its defining ideal. The following statement retains the marked quotient presentation.

Proposition 13.4 (A graded Hilbert locus and its first Betti space). *In the fixed coordinates of Theorem 7.3, let $S = \text{Sym } H^*$, $N = \binom{n+1}{2}$, $M = \binom{N}{2}$, and $Q_R = S/((K_R) + S_{\geq d+1})$. Then*

$$\text{Tor}_1^S(Q_R, \mathbb{C})_d = K_R. \quad (13.5)$$

This identity commutes with arbitrary complex base change. The functor of graded quotients with Hilbert function

$$h_j = \begin{cases} \dim S_j, & 0 \leq j < d, \\ \dim S_d - M, & j = d, \\ 0, & j > d \end{cases}$$

is represented by $\text{Gr}(M, S_d)$. The pencil locus is a closed copy of $\text{Gr}(2, \text{Sym}^2 V)$. Its degree- d first Betti space recovers every scheme-theoretic pencil parameter, including parameters over nonreduced bases.

Proof. Put $J = (K_R) + S_{\geq d+1}$ and $S_+ = \bigoplus_{j>0} S_j$. The augmented sequence $0 \rightarrow J \rightarrow S \rightarrow Q_R \rightarrow 0$ gives $\text{Tor}_1^S(Q_R, \mathbb{C}) = J/S_+J$, since its map to S/S_+ is zero. Its degree- d part is K_R . The same argument over a base algebra applies to the degree- d subbundle with locally free quotient and commutes with pullback. A graded quotient with the indicated locally free components has zero ideal below degree d , a rank- M subbundle in degree d , and the full ideal in higher degrees. Conversely, every such subbundle defines an ideal, as its positive-degree products lie above degree d . This is the Grassmannian functor. The closed immersion and its inverse on the image are Theorem 7.3. This is not an inference from closed-point Torelli to unmarked deformation-groupoid equivalence. $\square$

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